

Static probabilistic timing analysis of worst-case execution time for random replacement caches

Student: Bastien Pасdeloup

Supervisors: Isabelle Puaut, Damien Hardy

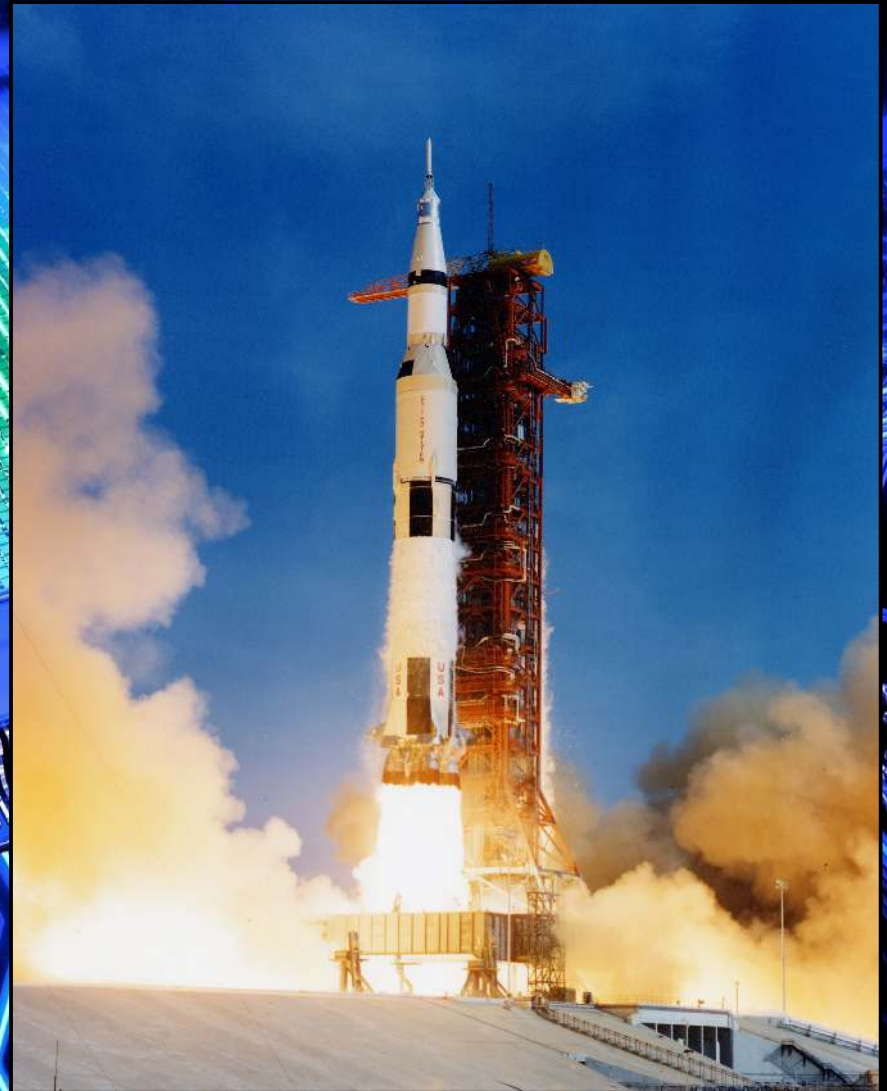
Research team: ALF

Wednesday, June 26th 2014

systems

pose method

Conc



Systems

Choose Method

Conc



Algorithms

Choose Method

Conc

Input domain

P

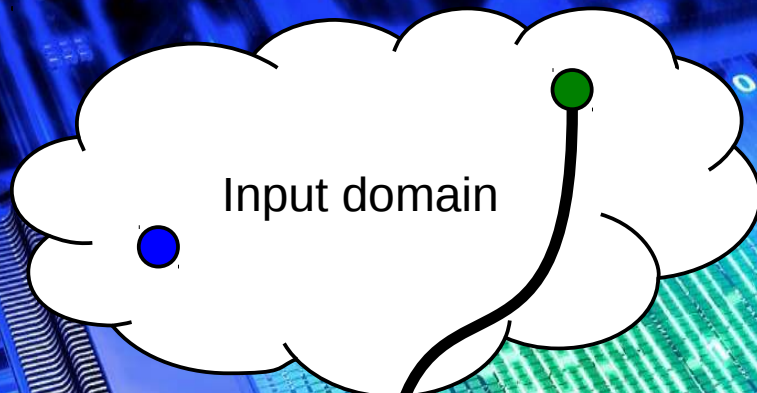
Output

T₁

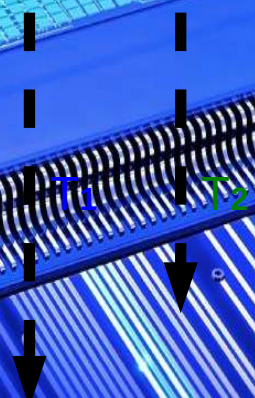
Algorithms

Choose Method

Conc



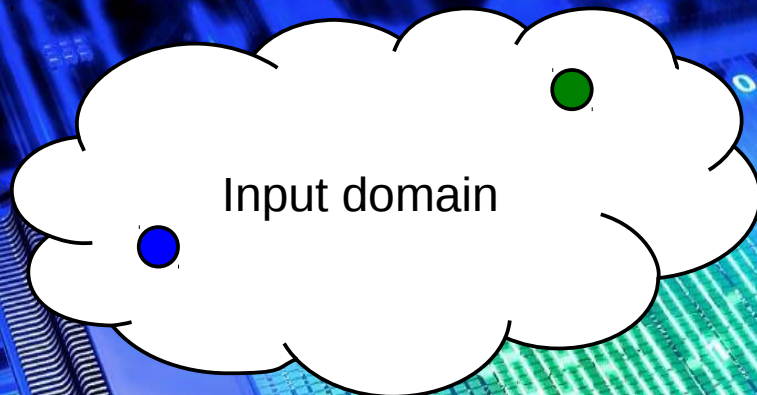
Output



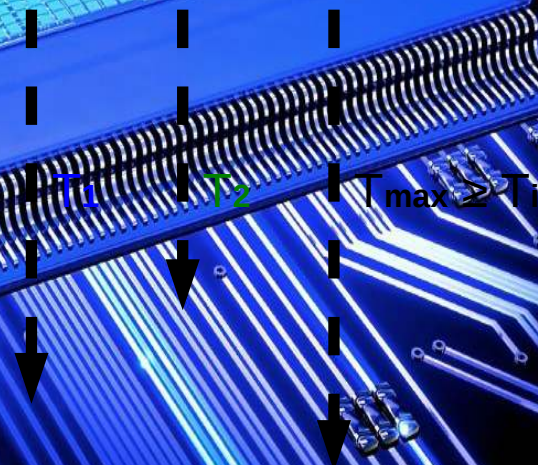
Algorithms

Choose Method

Conc



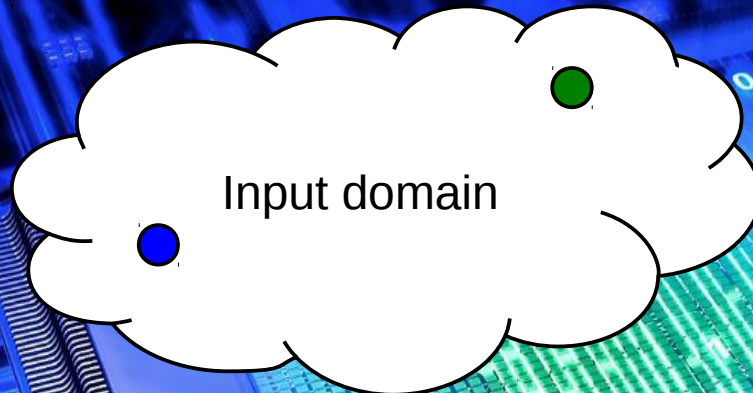
Output



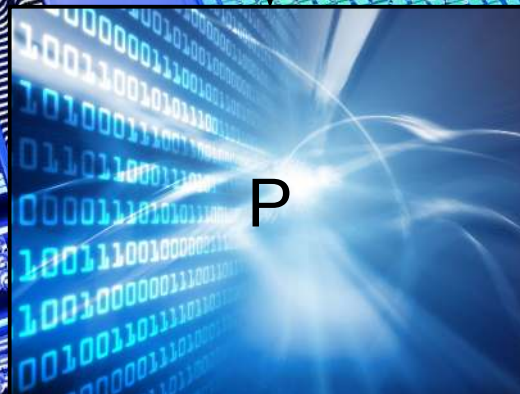
Algorithms

Choose Method

Conc



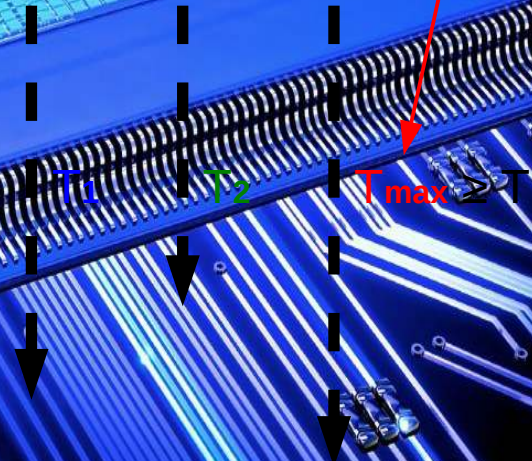
Input domain



P

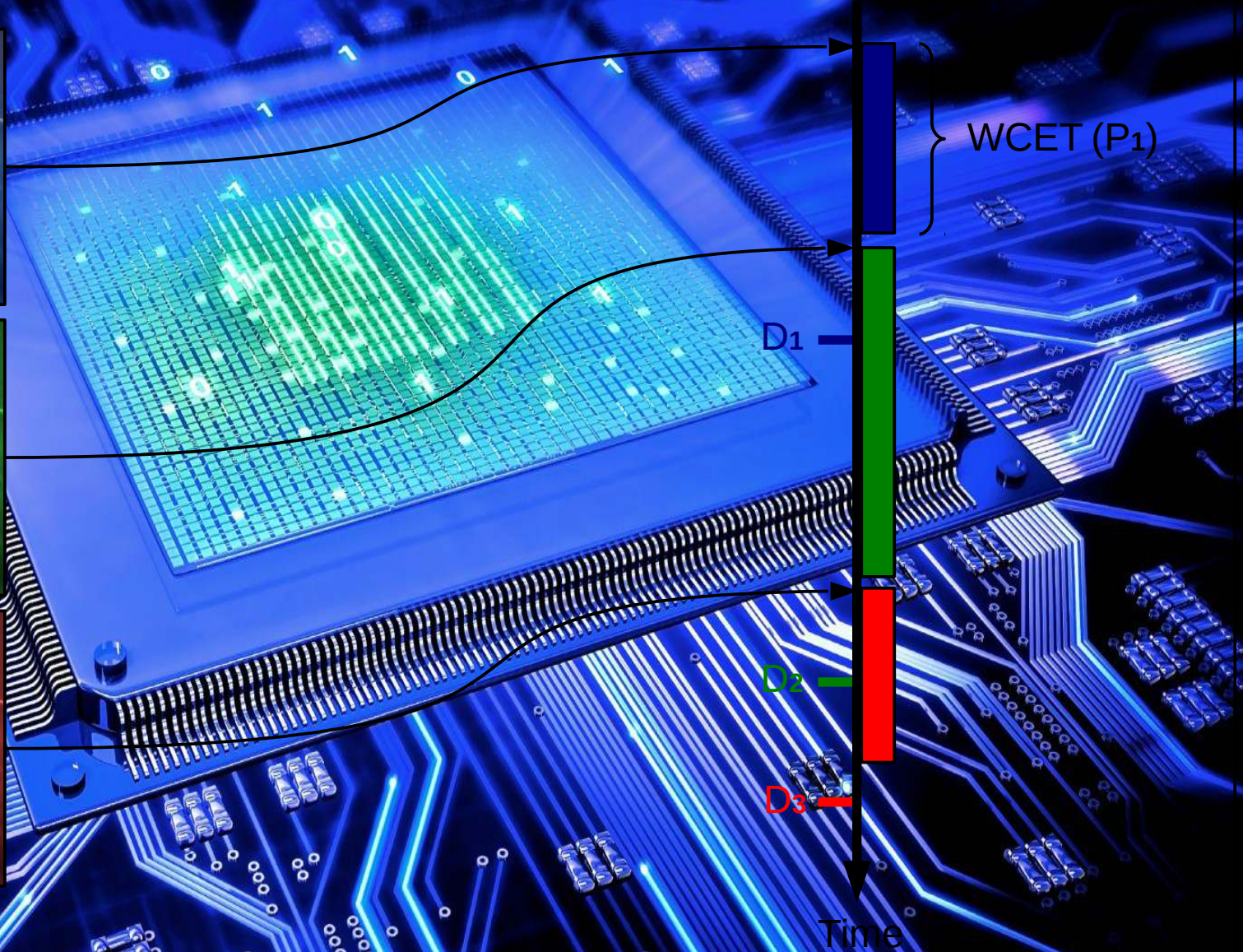
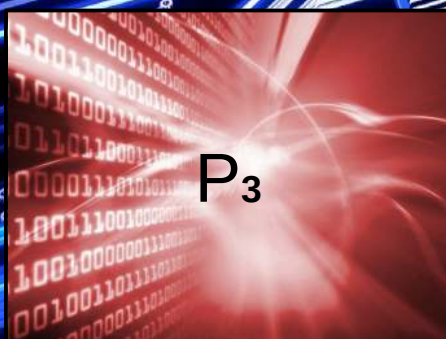
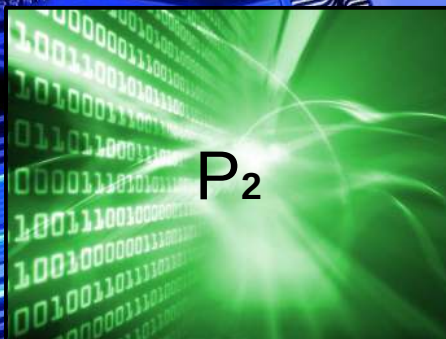
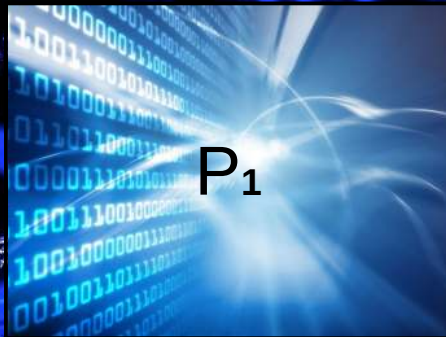
Output

Worst-case
Execution time
(WCET)



Programs

Choose Method
Conc



Conclusion

Proposed method

Conclusion

Safety

Estimate greater than the actual WCET

Tightness

As close as possible to the actual WCET

Actual
WCET

Execution
time

Measured execution times

Possible execution times

WCET
estimation
technique

WCET

Choose method

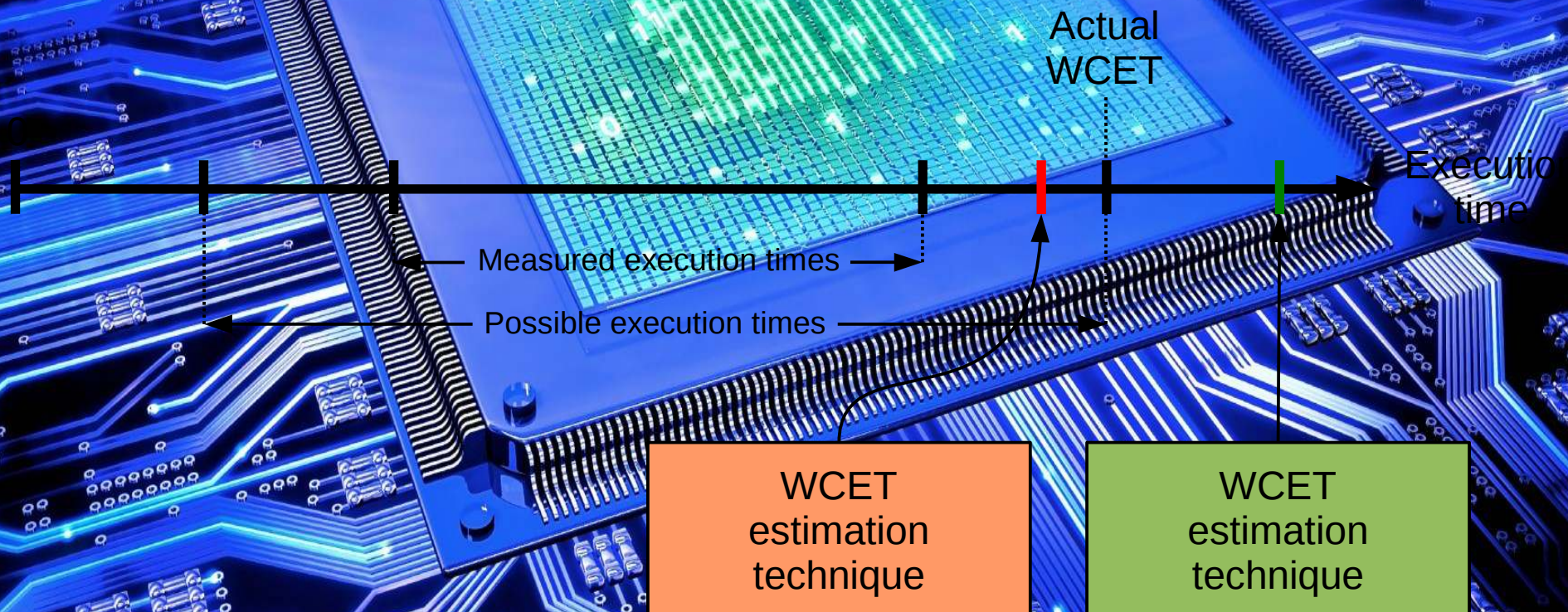
Conc

safety

Estimate greater than the actual WCET

Tightness

As close as possible to the actual WCET



Conclusion

Proposed method

Conclusion

Safety

Estimate greater than the actual WCET

Tightness

As close as possible to the actual WCET

Actual
WCET

Execution
time

Measured execution times

Possible execution times

WCET
estimation
technique

Robust

Proposed method

Conclusion

Deterministic techniques



**Deterministic
WCET estimation
technique**

Single WCET estimate

Probabilist

Proposed method

Conclusion

Deterministic techniques



Deterministic
WCET estimation
technique

Single WCET estimate

Probabilistic technique



Probabilistic
WCET estimation
technique

Execution time pr
(timing values + probabilities)

Probabilistic

Proposed method

Conclusion

Deterministic techniques



**Deterministic
WCET estimation
technique**

Single WCET estimate

Probabilistic technique

**Probabilistic
WCET estimation
technique**

**Execution time profile
(timing values + probabilities)**

Probability of
exceedance

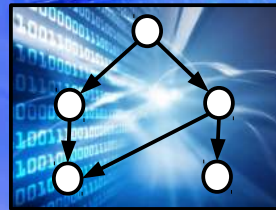


Simulation-based

Proposed method

Conclusion

Static techniques



Low-level
analysis

High-level
analysis

WCET estimation technique

Result

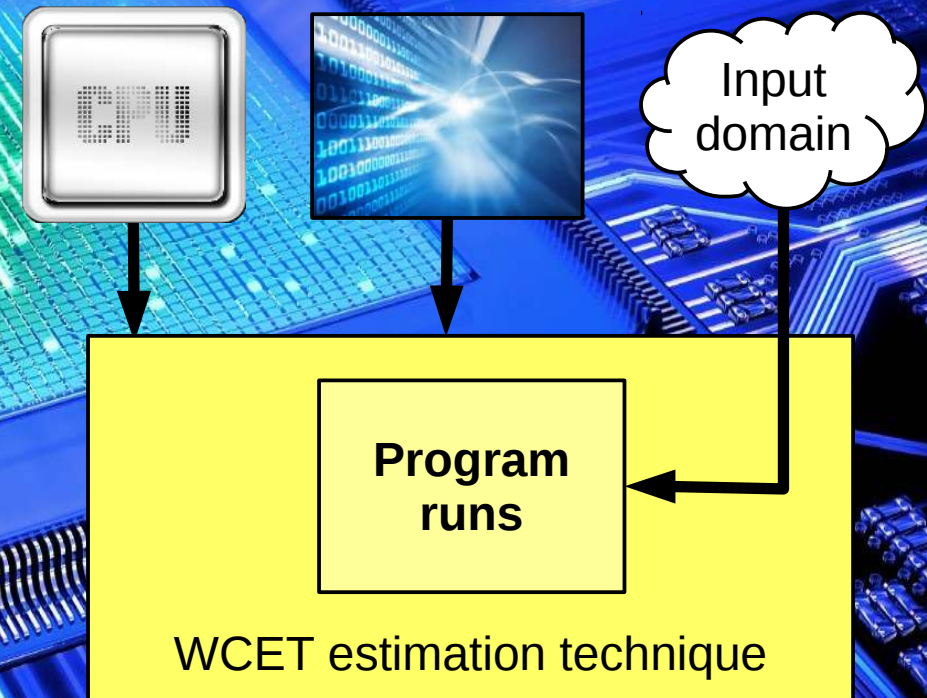
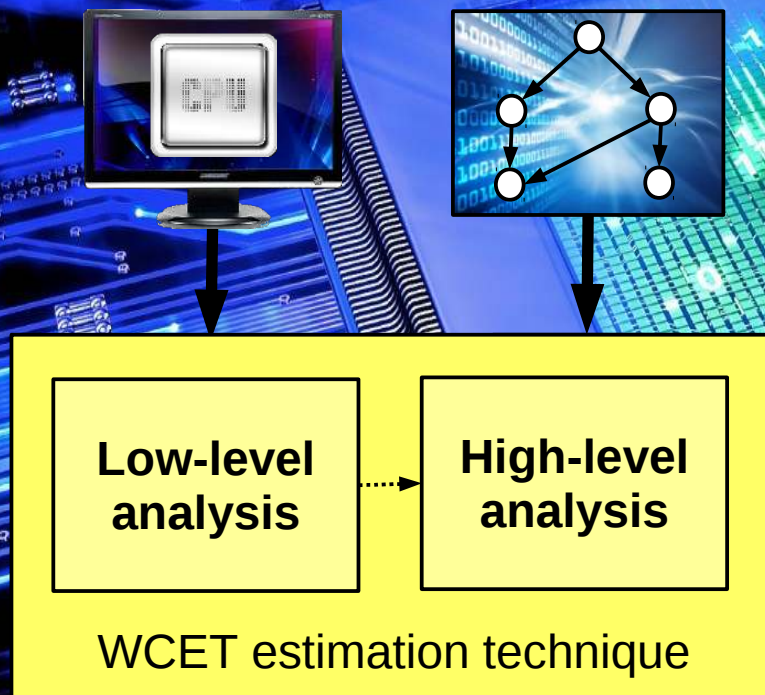
Static-based

Proposed method

Conclusion

Static techniques

Measurement-based techniques



Methods

Choose method
Conclude

		Deterministic	Probabilistic
Static methods	High-level	[Pus1989] [Ste1995]	[Dav2004]
	Low-level	[Fer1999]	[Alt2014]
Measurement-based methods		[Wil2005] [Wil2008]	[Ber2002] [Cuc2012]

- [Alt2014] S. Altmeyer, R. I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis" (2014)
- [Ber2002] G. Bernat, A. Colin, S.M. Petters – "WCET Analysis of Probabilistic Hard Real-Time Systems" (2002)
- [Cuc2012] L. Cucu-Grosjean et. al – "Measurement-Based Probabilistic Timing Analysis for Multi-path Programs" (2012)
- [Dav2004] J. David, J. Puaut – "Static Determination of Probabilistic Execution Times" (2004)
- [Fer1999] C. Ferdinand, R. Wilhelm – "Efficient and Precise Cache Behavior Prediction for Real Time Systems" (1999)
- [Pus1989] P. Puschner, C. Koza – "Calculating the Maximum Execution Time of Real-Time Programs" (1989)
- [Ste1995] Y.T. Steven Li, S. Malik – "Performance Analysis of Embedded Software Using Implicit Path Enumeration" (1995)
- [Wil2008] R. Wilhelm et. al – "The Worst-case Execution-time Problem – Overview of Methods and Survey of Tools" (2008)
- [Wil2005] R. Williams et. al – "PathCrawler: automatic generation of path tests by combining static and dynamic analysis" (2005)

Methods

Propose Method
Conclude

		Deterministic	Probabilistic
Static methods	High-level	[Pus1989] [Ste1995]	[Dav2004]
	Low-level	[Fer1999]	[Alt2014]
Measurement-based methods		[Wil2005] [Wil2008]	[Ber2002] [Cuc2012]

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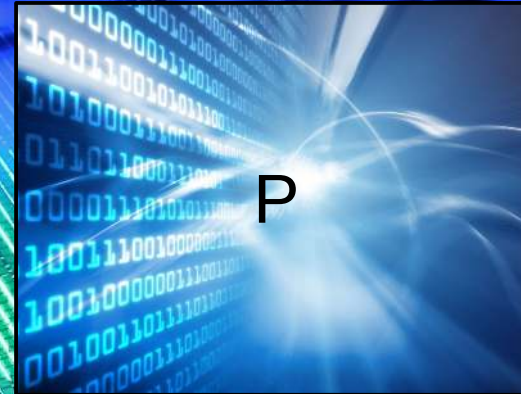
Static
[Alt2014]

Dynamic
Chosen method
Conc

- ▶ Probabilistic
- ▶ Static
- ▶ Low-level
- ▶ Single-path

[Alt2014] Alt, R. I. Davis — "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis"

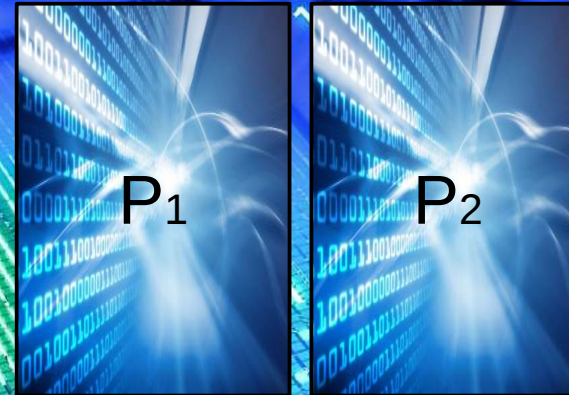
- ▶ Static
- ▶ Low-level
- ▶ Single-path



Static
[Alt2014]

Choose method
Conc

- ▶ Probabilistic
- ▶ Static
- ▶ Low-level
- ▶ Single-path



Exhaustive analysis
(precise & long)

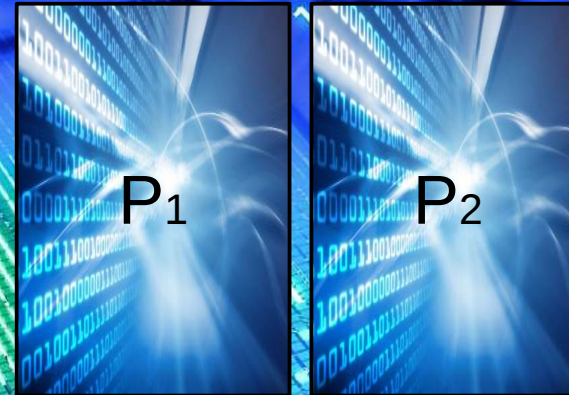
Formula
(imprecise & fast)

[Alt2014] Bastien P. Meyer, R.I. Davis — "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis"

Static
[Alt2014]

Probabilistic
Chosen method
Conc

- ▶ Probabilistic
- ▶ Static
- ▶ Low-level
- ▶ Single-path



Exhaustive analysis
(precise & long)

Formula
(imprecise & fast)

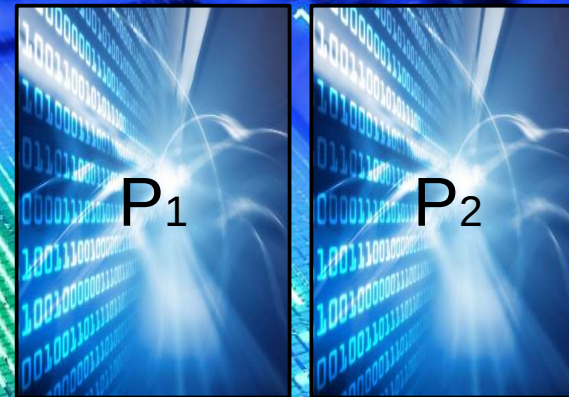
Distribution of
probabilities

[Alt2014] Bastien P. Meyer, R.I. Davis — "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis"

Static Probabilistic Timing Analysis [Alt2014]

Qualitative
Quantitative
Choose method
Conclusion

- Probabilistic
- Static
- Low-level
- Single-path



Exhaustive analysis
(precise & long)

Formula
(imprecise & fast)

Feasible?

Distribution of
probabilities

[Alt2014] Bastien P. Meyer, R.I. Davis — "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis"

- Probabilistic
- Static
- Low-level
- Single-path



Exhaustive analysis
(precise & long)

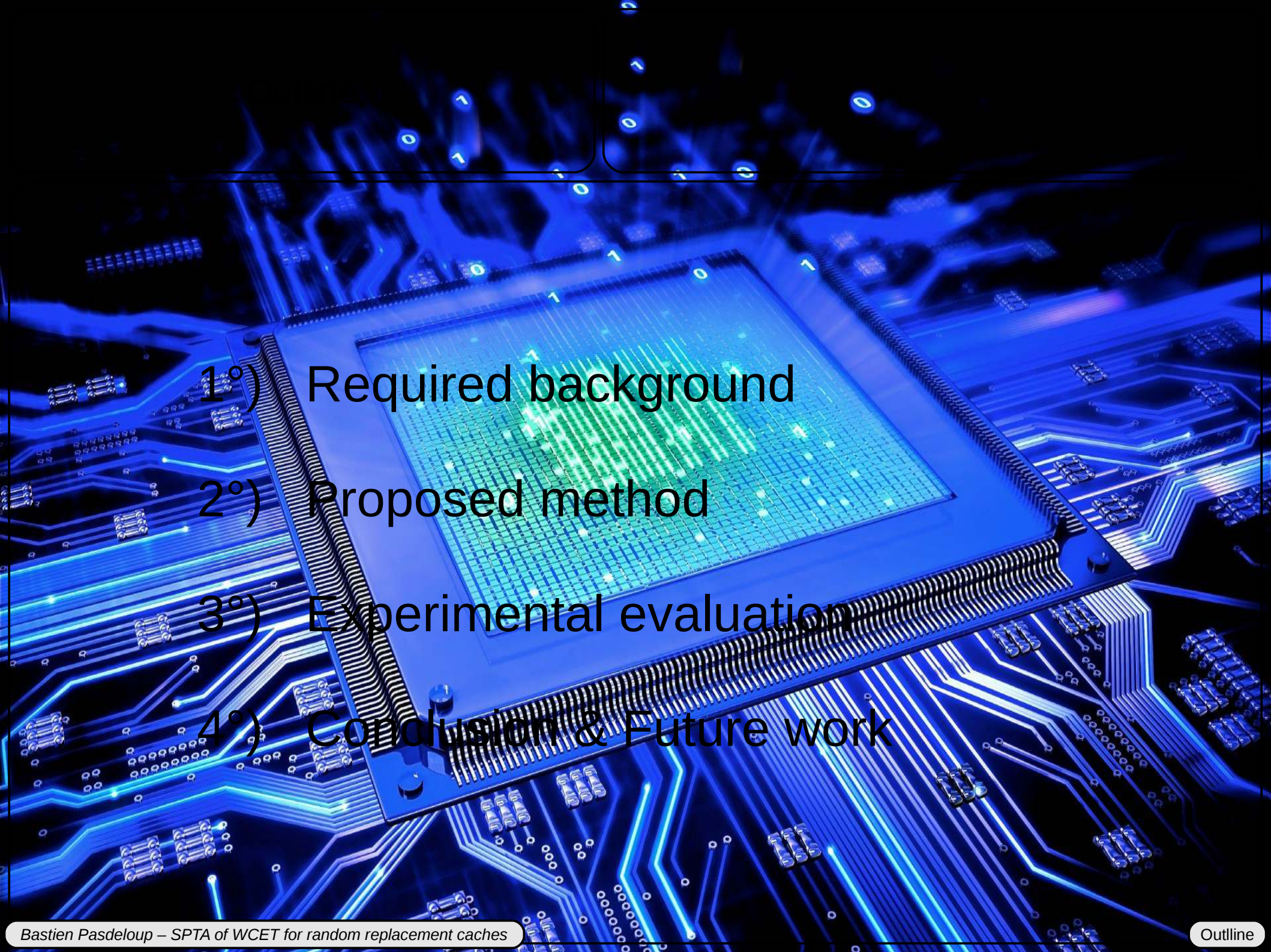
Formula
(imprecise & fast)

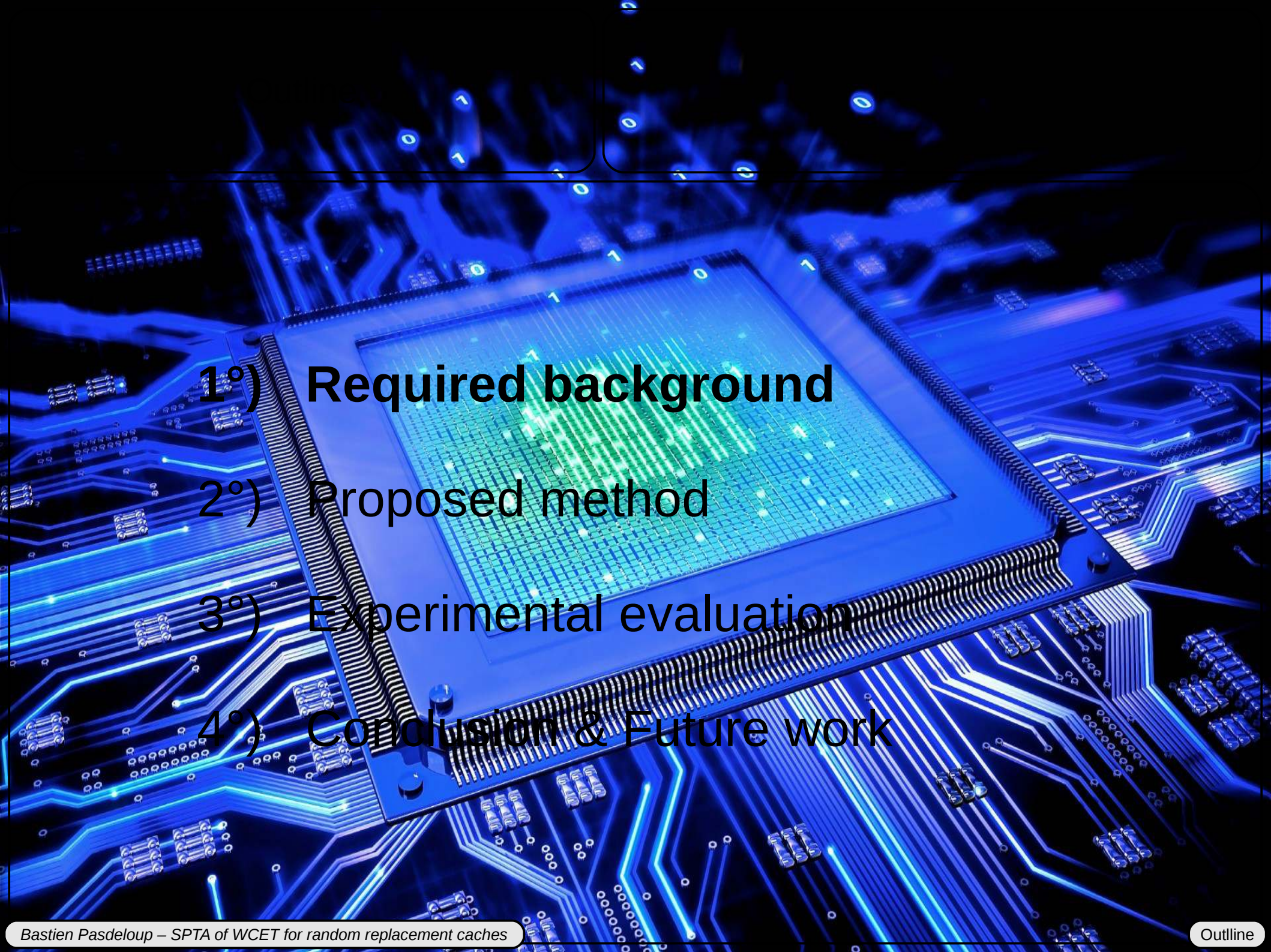
Objectives:

- Same context
- Safe & tighter

Feasible?

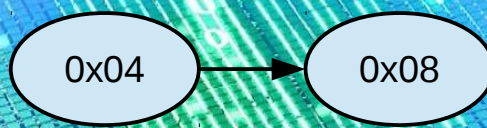
Distribution of
probabilities

- 
- 1°) Required background
 - 2°) Proposed method
 - 3°) Experimental evaluation
 - 4°) Conclusion & Future work

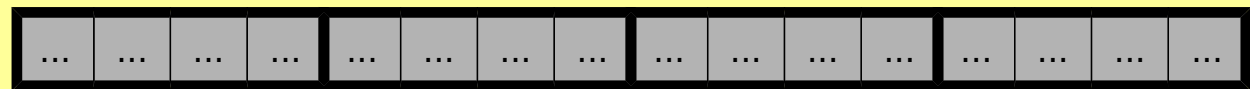
- 
- 1°) Required background
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 - 4°) Conclusion & Future work

• Small, fast memories
 • Used in most architectures
 • Takes advantage of the locality

• First-in, first-out (FIFO)
 • Least-recently-used (LRU)
 • Random
 • Choose method
 • Concurrent



Cache
 (~2 cycles)
 Main memory
 (~70 cycles)



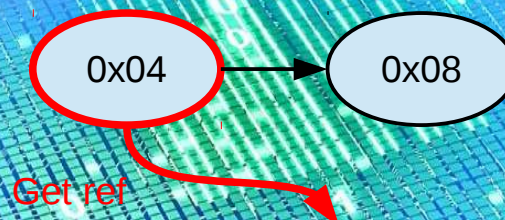
Replacement policy

00	04	08	0c	10	14	18	1c	20	24	28	2c	30	34	38	3c	40	44	48	4c	50	54	58	5c
60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...

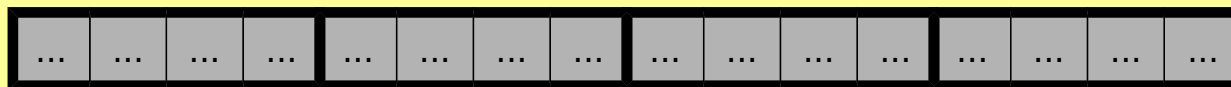
• Cache coherence
 • Cache consistency
 • Cache non-coherence

• Cache replacement
 • Cache size
 • Cache access method
 • Cache access latency
 • Cache access bandwidth
 • Cache access cost

- Small, fast memories
- Present in most architectures
- Takes advantage of the locality



Cache
 (~2 cycles)
 Main memory
 (~70 cycles)



Replacement policy

00	04	08	0c	10	14	18	1c	20	24	28	2c	30	34	38	3c	40	44	48	4c	50	54	58	5c
60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...

Cache
(~2 cycles)

Main memory
(~70 cycles)

Miss

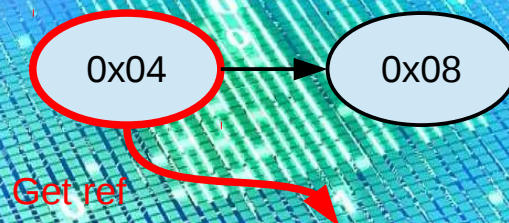
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
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Replacement policy

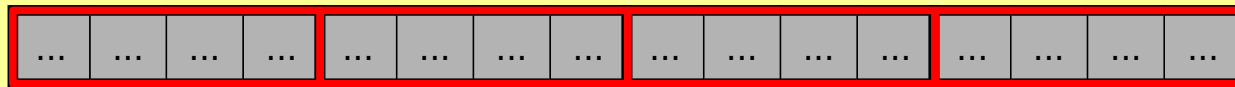
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60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...



- Small, fast memories
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Miss



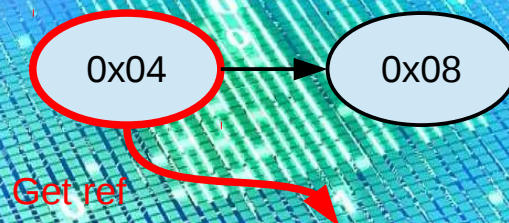
Replacement policy



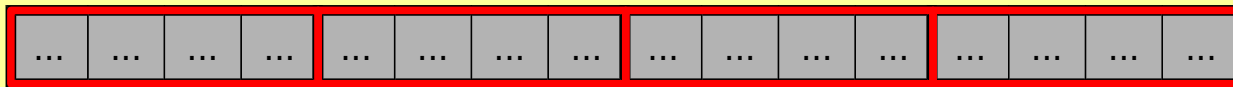
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60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...



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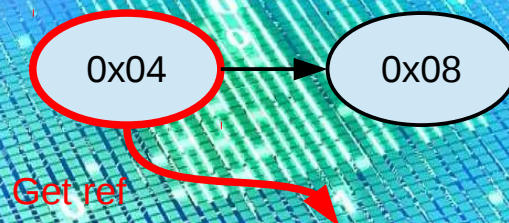
Replacement policy



00	04	08	0c	10	14	18	1c	20	24	28	2c	30	34	38	3c	40	44	48	4c	50	54	58	5c
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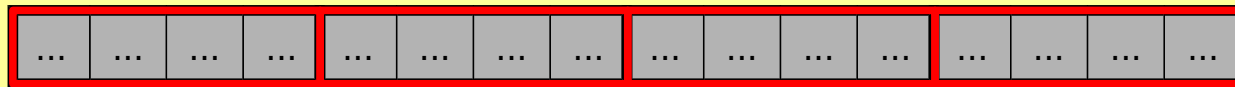
- Small, fast memories
- Present in most architectures
- Takes advantage of the locality



Cache
(~2 cycles)

Main memory
(~70 cycles)

Miss



Replacement
policy

Update way

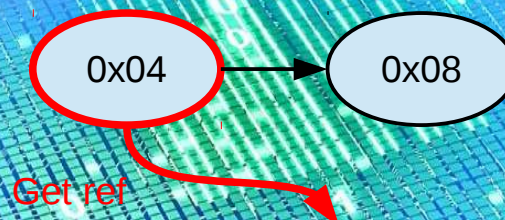


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60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...

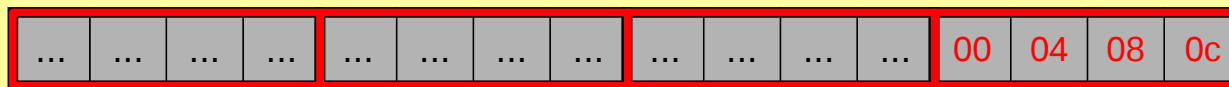
Small, fast memories
in most architectures

Choose method
Conc

- Small, fast memories
- in most architectures
- Takes advantage of the locality



Miss



Replacement
policy

Update way

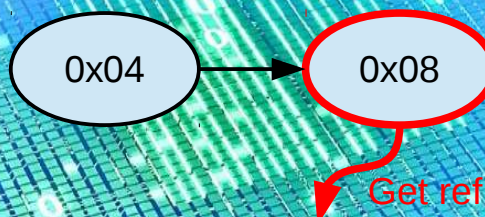
Get ref

00	04	08	0c	10	14	18	1c	20	24	28	2c	30	34	38	3c	40	44	48	4c	50	54	58	5c
60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...

Small, fast memories
in most architectures

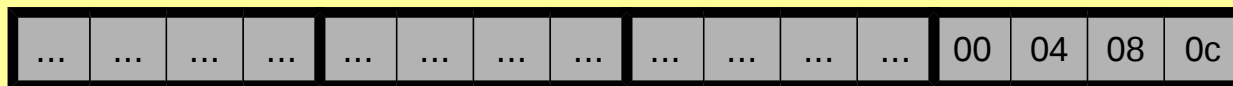
Choose method
Conc

- Small, fast memories
- in most architectures
- Takes advantage of the locality



Cache
(~2 cycles)

Main memory
(~70 cycles)



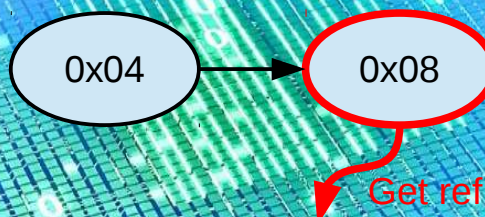
Replacement
policy

00	04	08	0c	10	14	18	1c	20	24	28	2c	30	34	38	3c	40	44	48	4c	50	54	58	5c
60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...

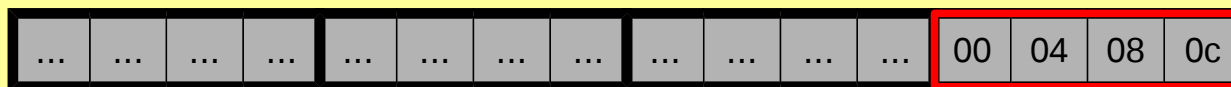
Small, fast memories
in most architectures

Choose method
Conc

- Small, fast memories
- in most architectures
- Takes advantage of the locality

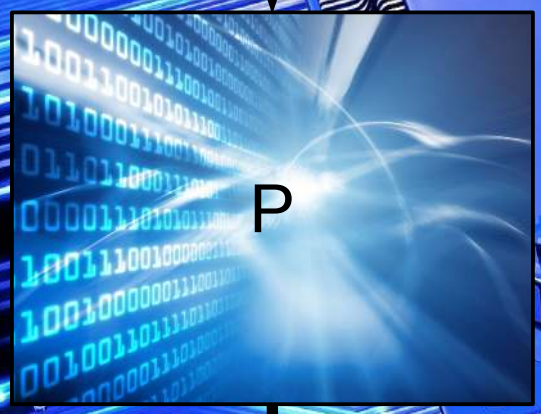
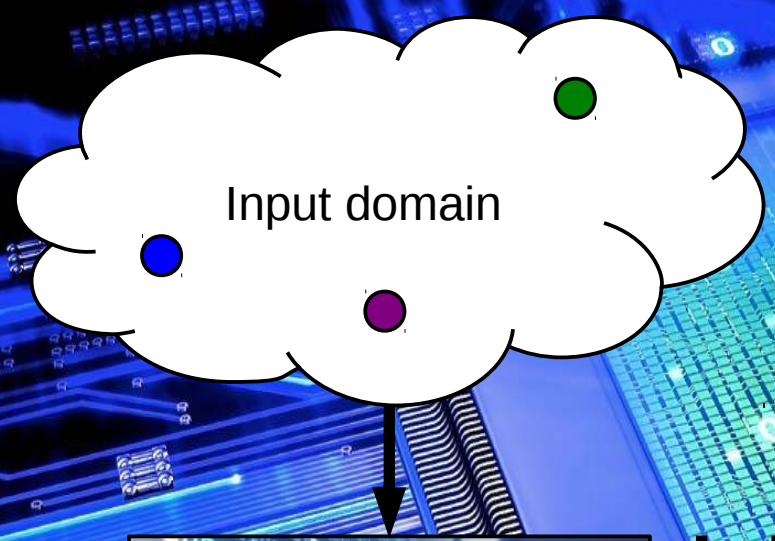


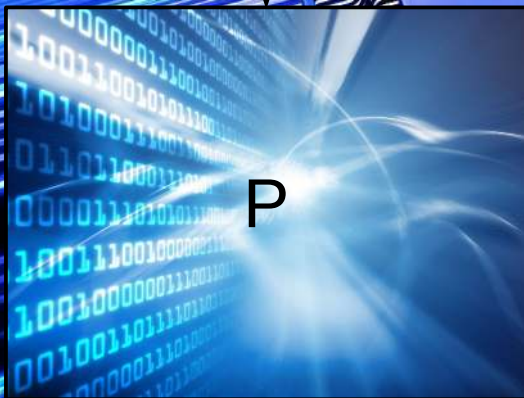
Hit



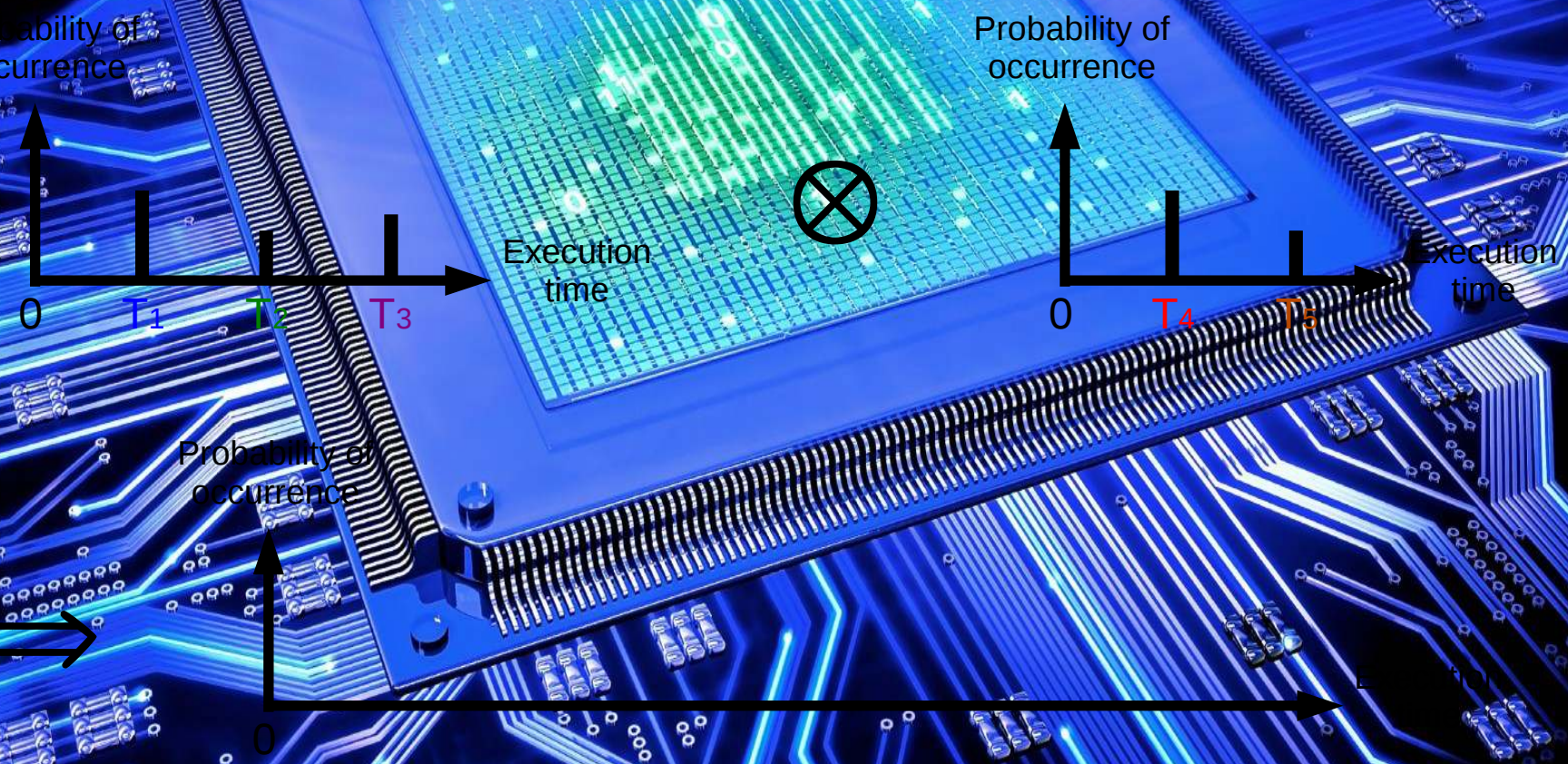
Replacement
policy

00	04	08	0c	10	14	18	1c	20	24	28	2c	30	34	38	3c	40	44	48	4c	50	54	58	5c
60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...





independence between distributions



Proposed Method

Proposed Method

Conclusion

independence between distributions

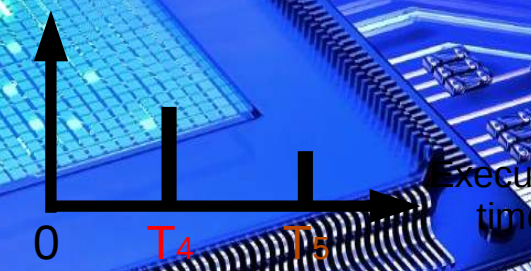
Probability of occurrence



Execution time



Probability of occurrence



Execution time

Probability of occurrence



P_4

Proposed Method

Proposed Method

Conclusion

independence between distributions

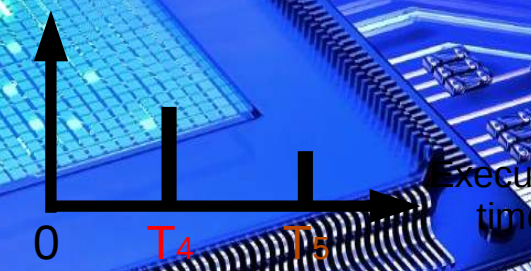
Probability of occurrence



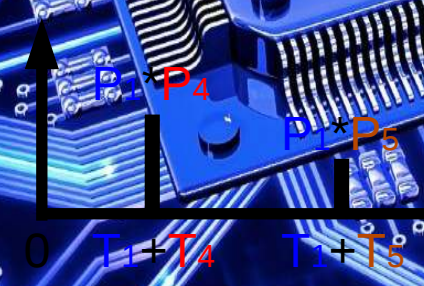
Execution time



Probability of occurrence



Probability of occurrence



Proposed Method

Proposed Method

Conclusion

independence between distributions

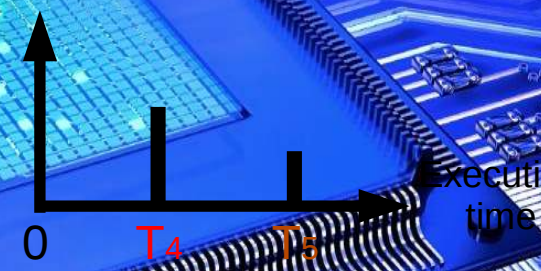
Probability of occurrence



Execution time



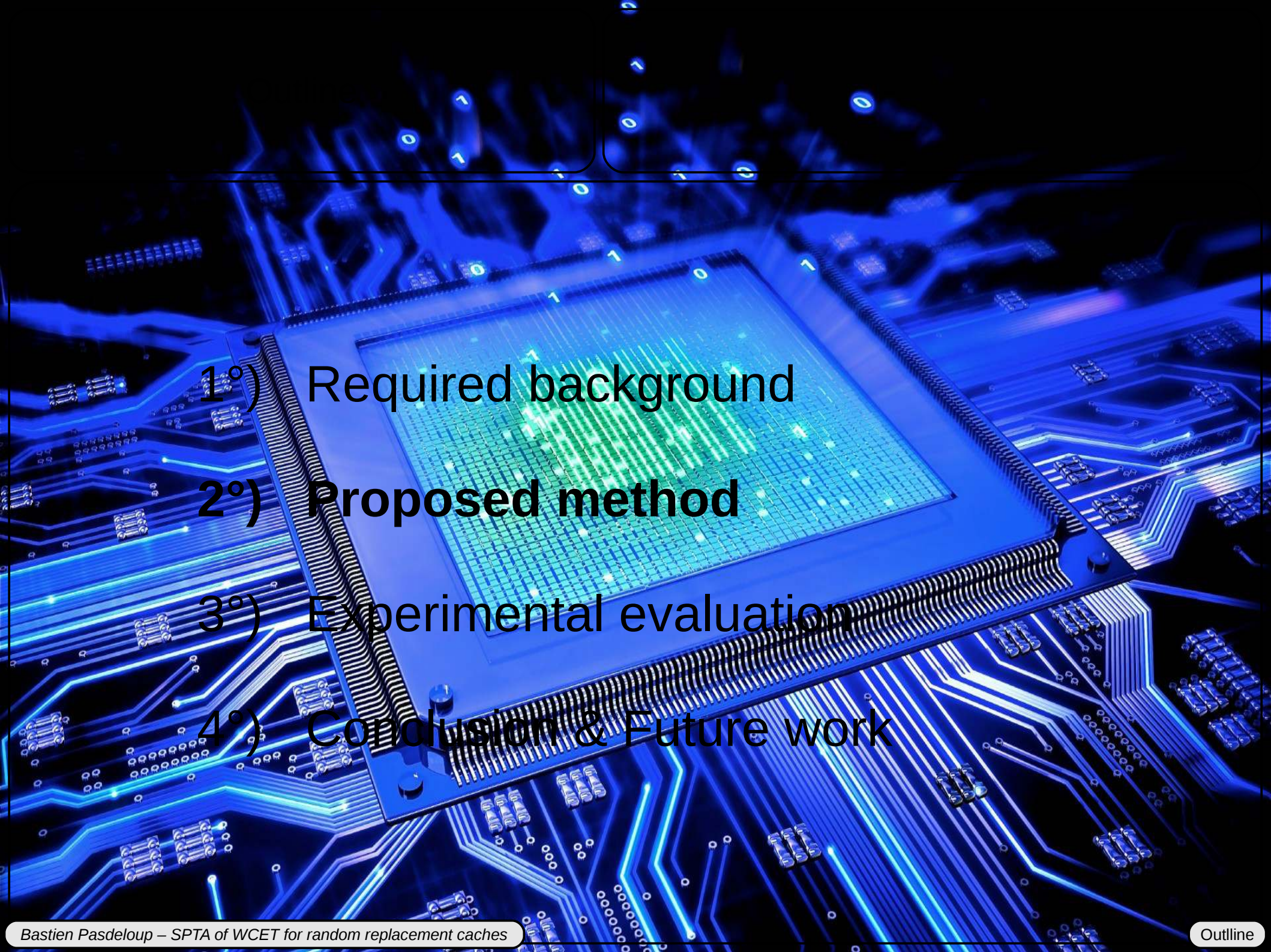
Probability of occurrence



Execution time

Probability of occurrence



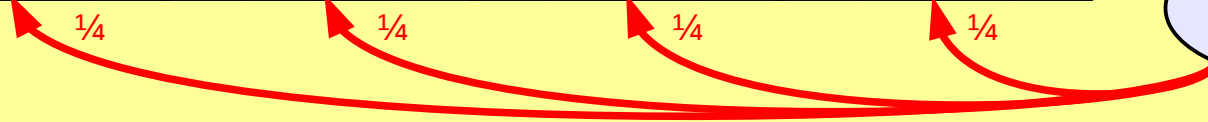
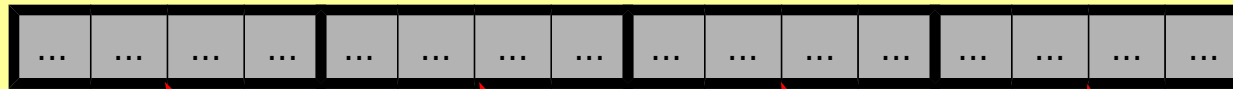
- 
- 1°) Required background
 - 2°) **Proposed method**
 - 3°) Experimental evaluation
 - 4°) Conclusion & Future work

software

used method

Conc

replacement caches (4 ways)



Replacement
policy

Caching



```

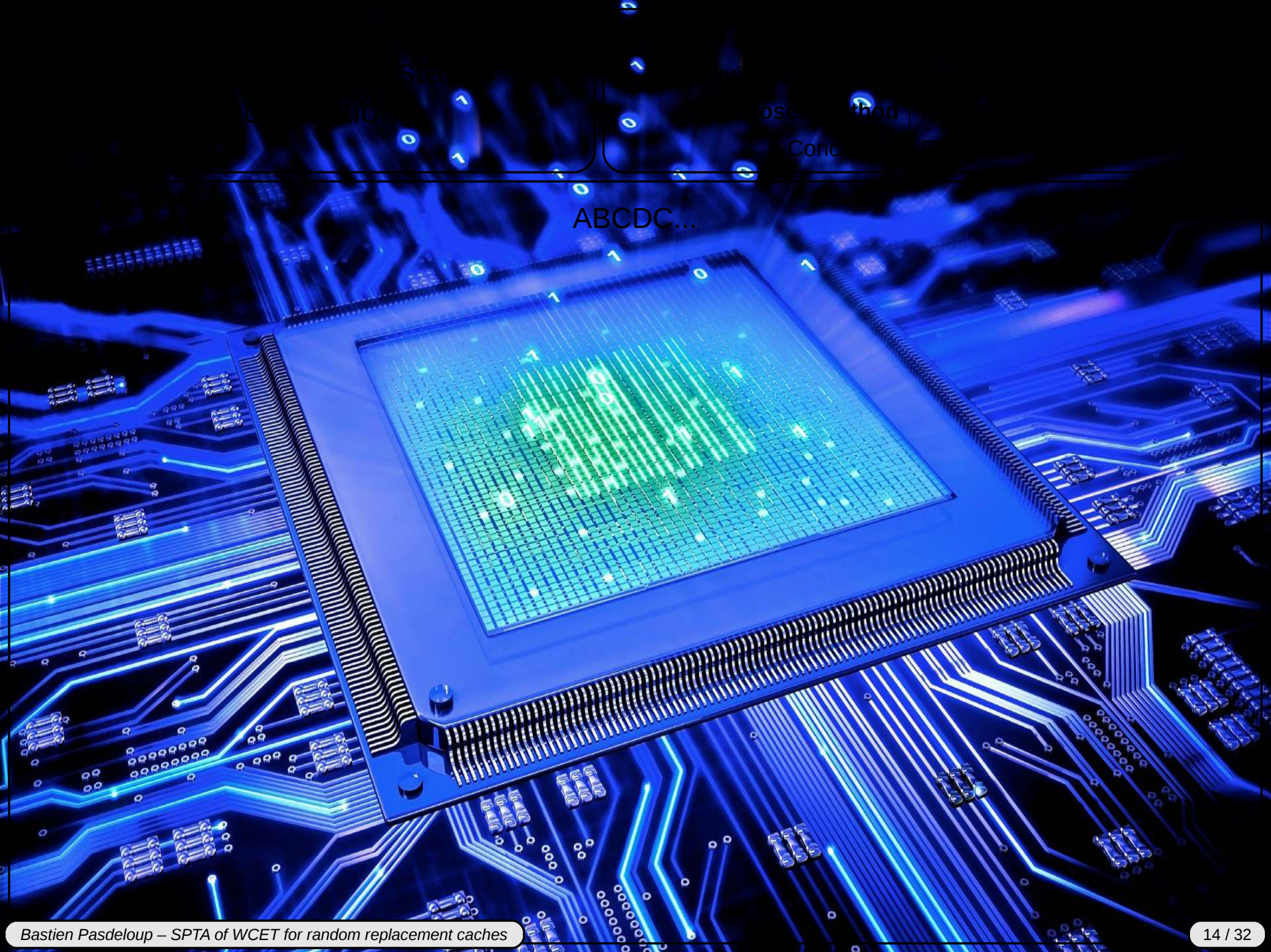
graph LR
    A((A)) -- 1 --> B((B))
    B -- 2 --> C((C))
    C -- 1 --> D((D))
    D -- 1 --> E((E))
    E -- 1 --> F((F))
    E -- 3 --> C
    D -- 4 --> B
  
```

Bastien Paskdeloup – SPTA of WCET for random replacement caches

Static
Uncertainty

Dynamic
Proposed Method
Concurrence

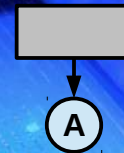
ABCD...C...



State
Transition

Analysis
Method
Conc.

(A)BCDC...



State
Transition

Proposed Method
Concurrence

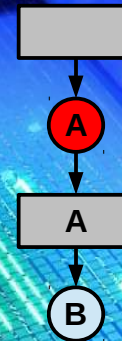
(A)BCDC...



State
Transition

Analysis
Method
Concurrence

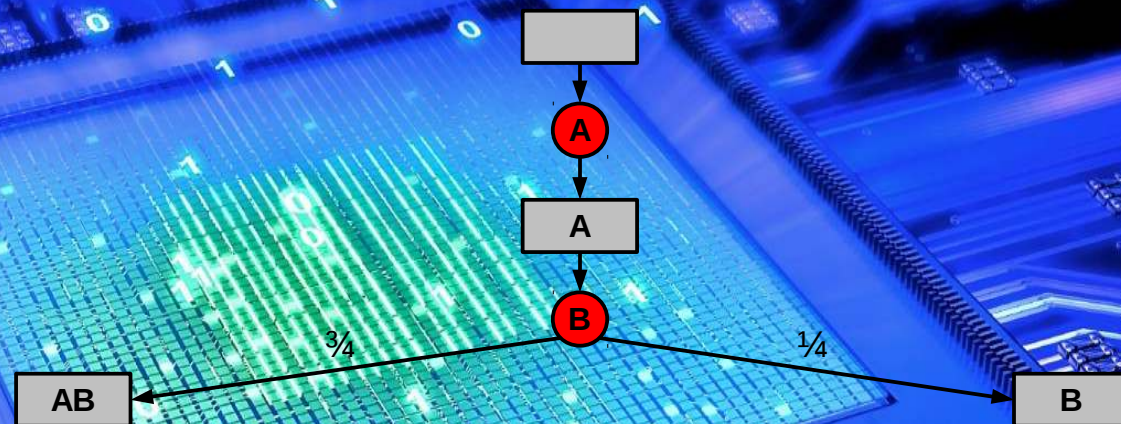
A(B)CDC...



State
Transition

Analysis
Method
Conc.

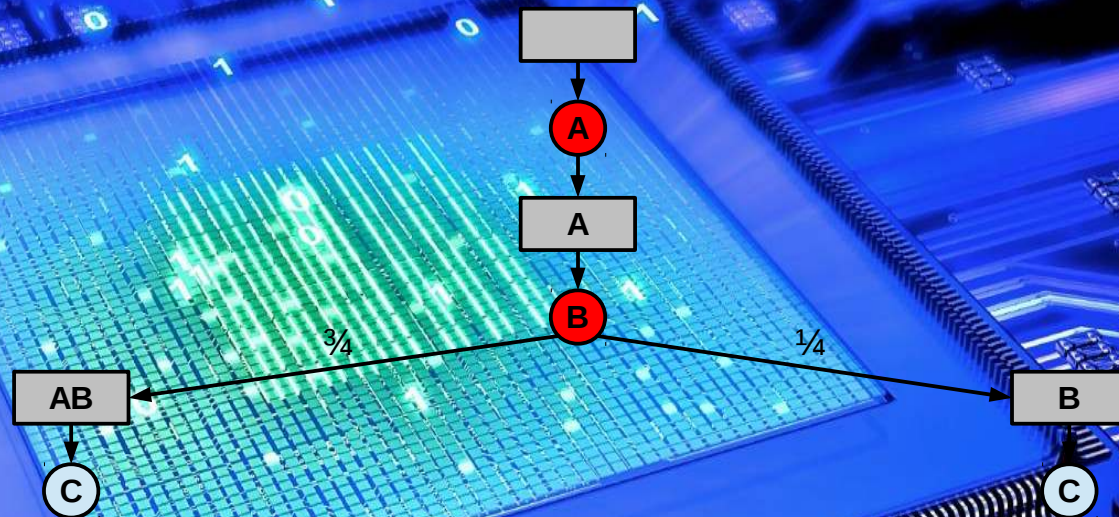
A(B)CDC...



State
Transition

Proposed Method
Concurrence

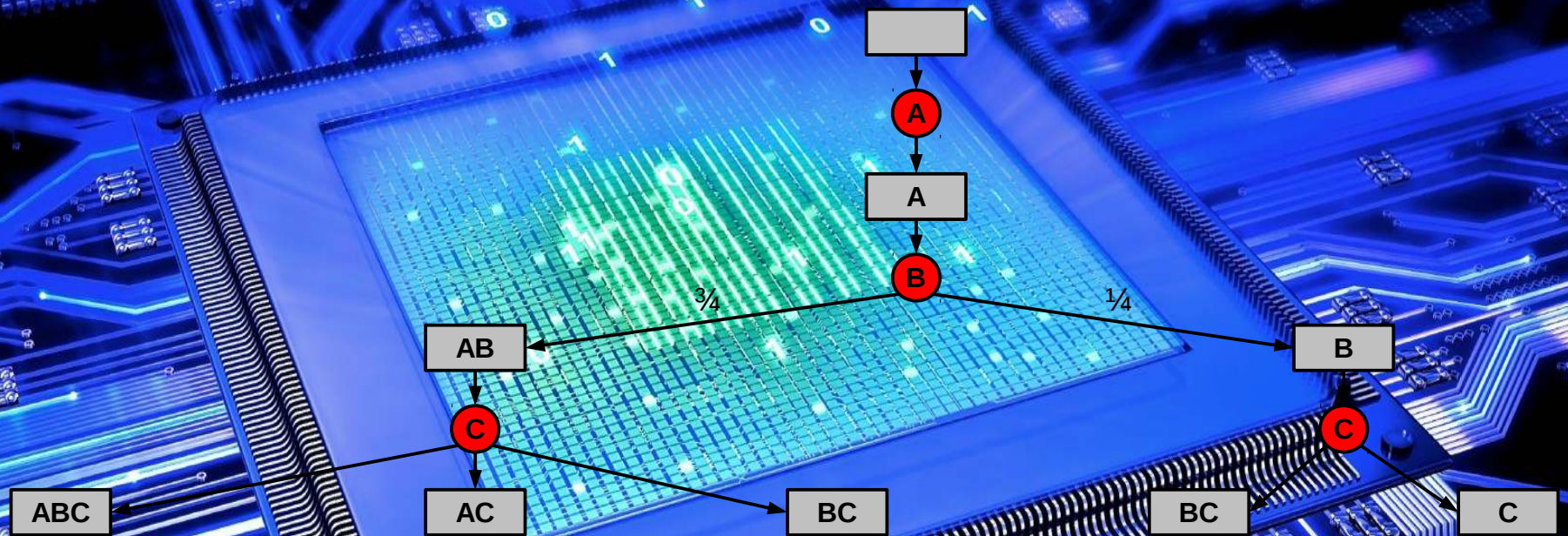
AB(C)DC...



State
Transition

Proposed Method
Concurrence

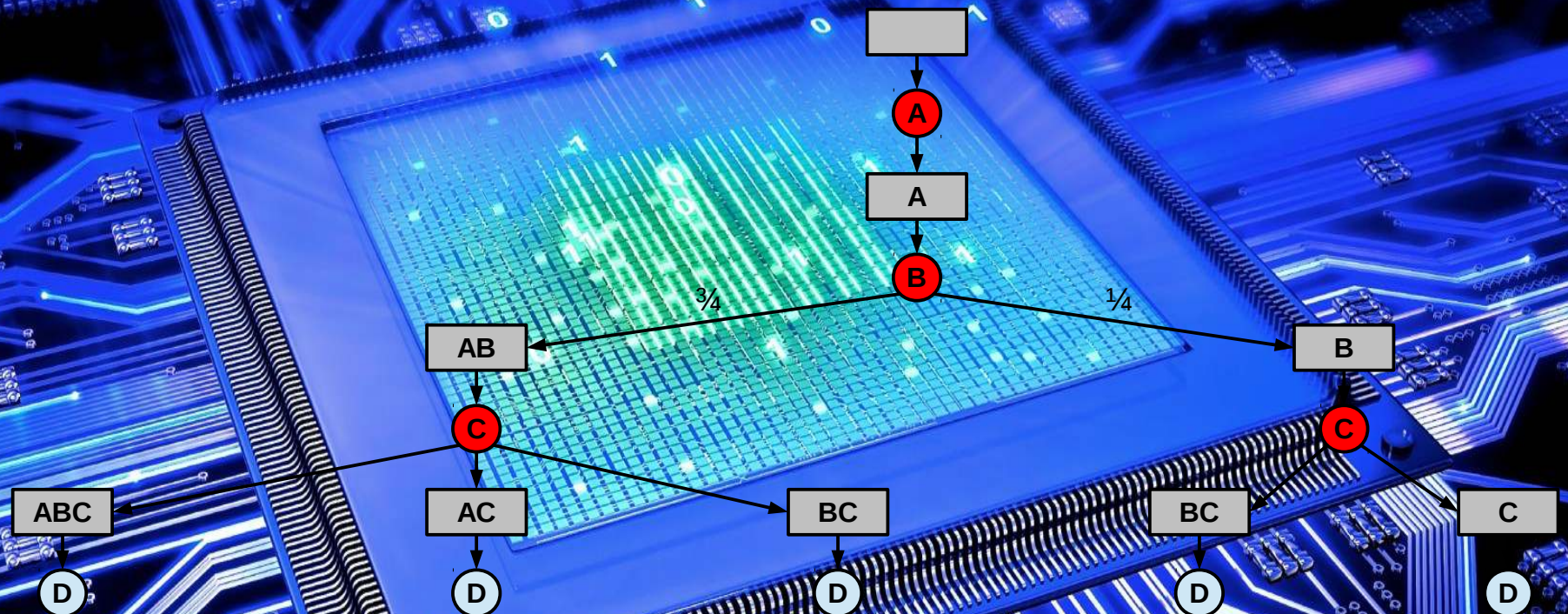
AB(C)DC...



State
Transition

Proposed Method
Concurrence

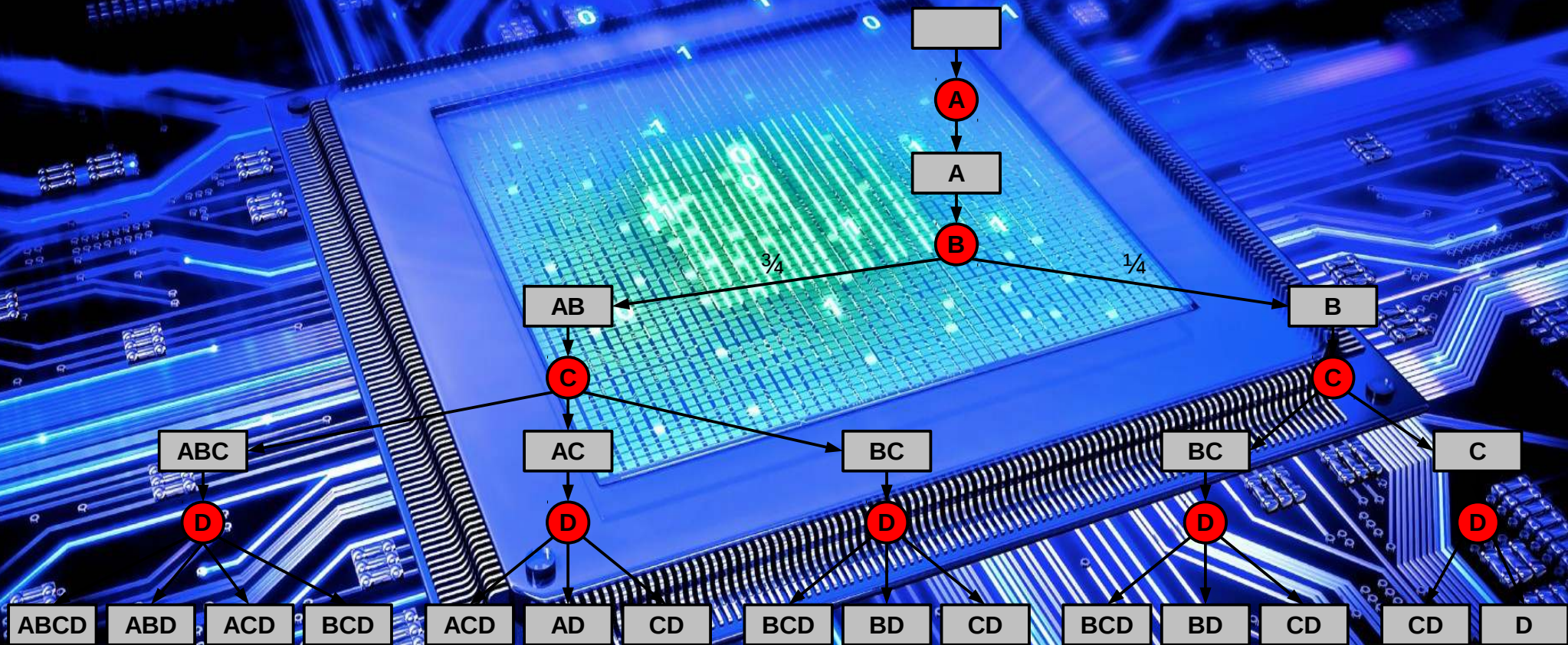
ABC(D)C...



State
Transition

Proposed Method
Concurrence

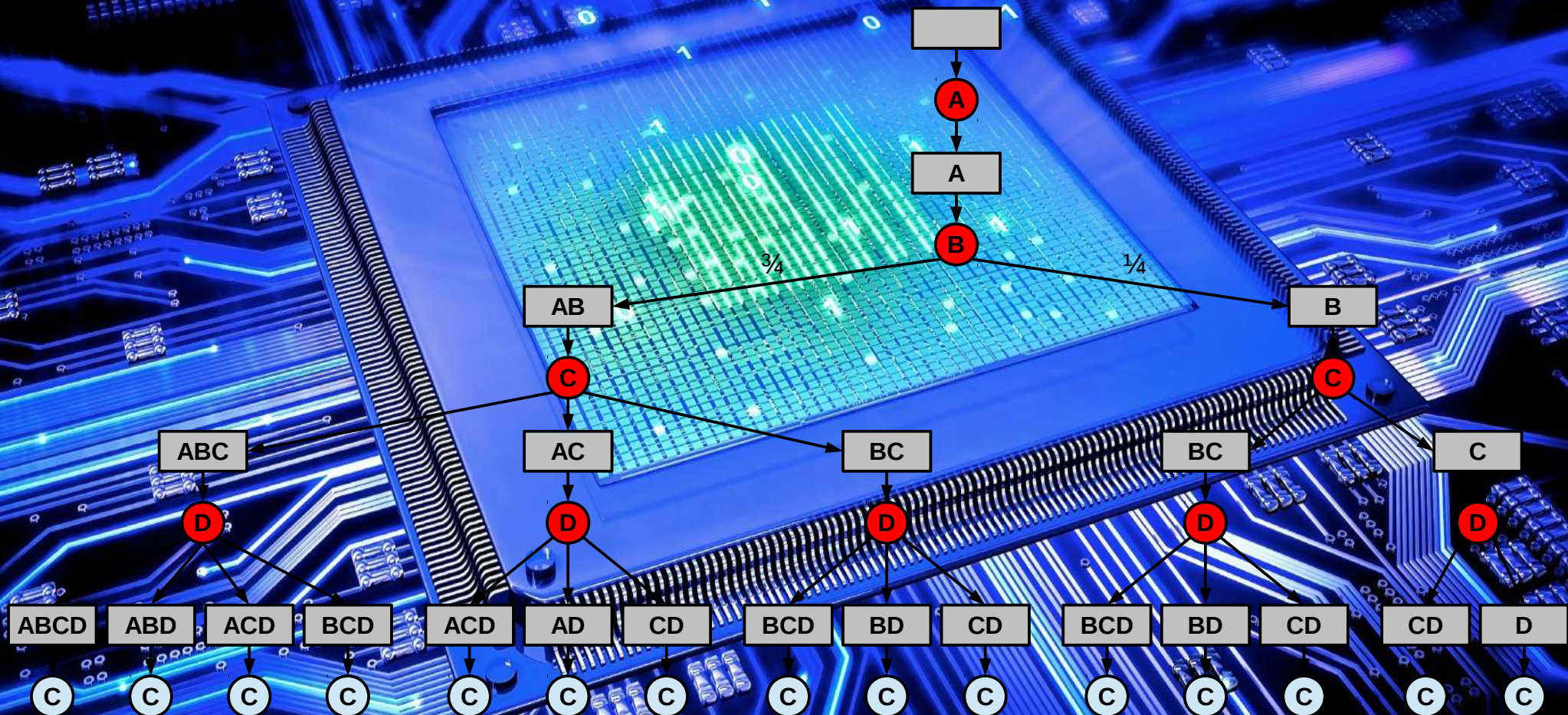
ABC(D)C...



State
Transition

Access Method
Concurrence

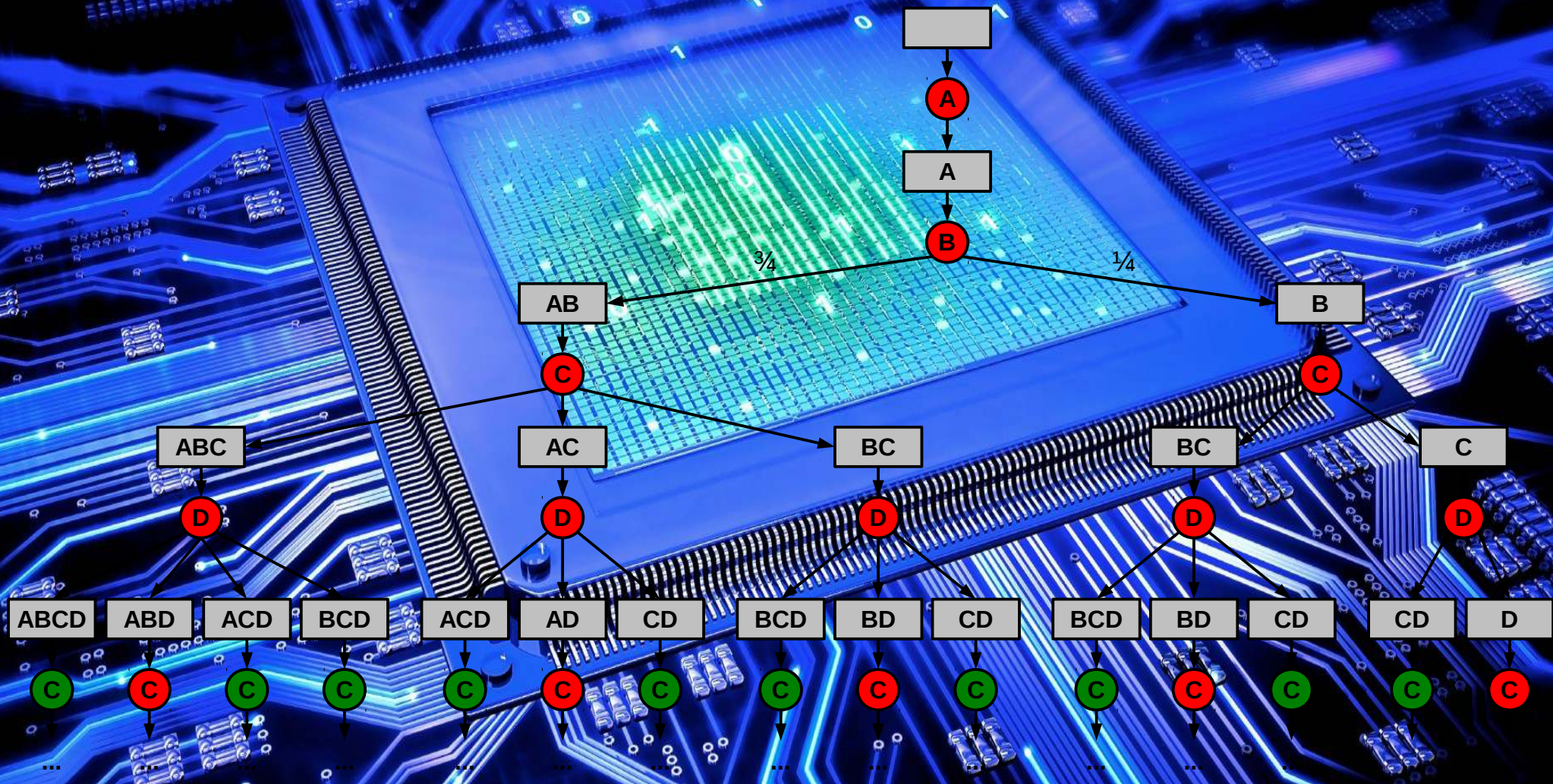
ABCD(C)...



State
Transition

Proposed Method
Concurrence

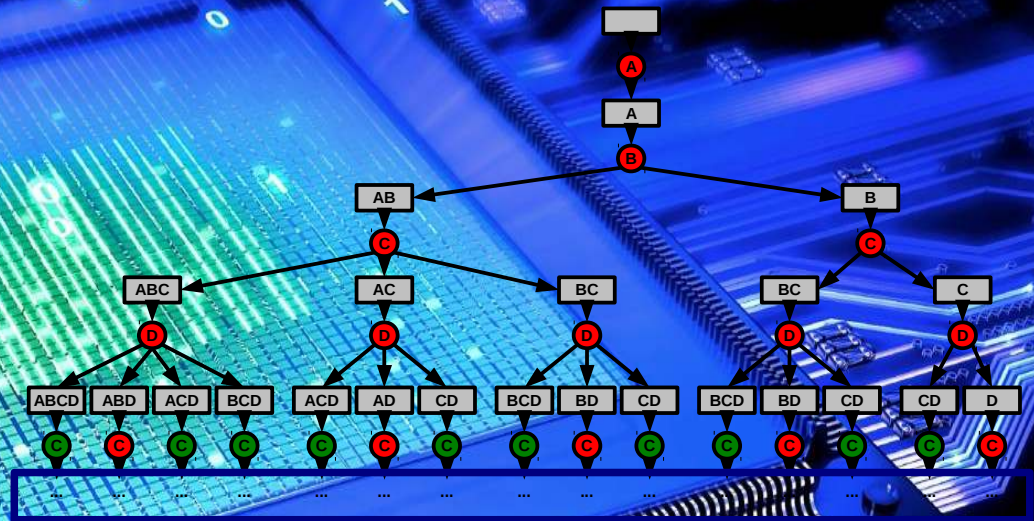
ABCD(C)...



the probability
of occurrence

the proposed method
Concurrence

Probability of
occurrence



Execution
time

0 (α^H, β^M) (α'^H, β'^M) (α''^H, β''^M) ...

Proposed method

Concave



- ▶ Exponential complexity
- ▶ Intractable

- ▶ Real probabilities
- ▶ Exact distribution

Proposed method

Concave



- ▶ Exponential complexity
- ▶ Intractable

- ▶ Real probabilities
- ▶ Exact distribution

Idea:

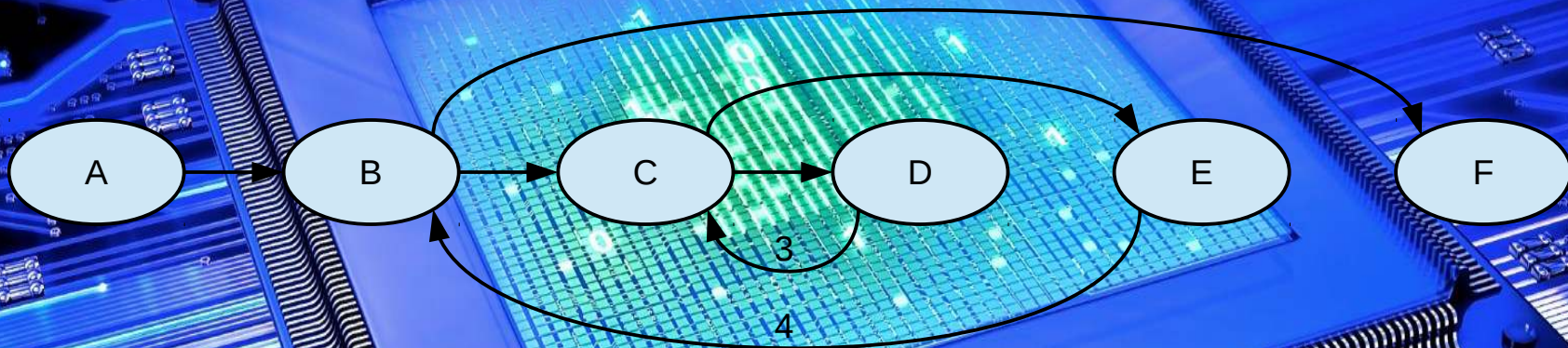
- ▶ Cut into windows
- ▶ Exhaustive analyses
- ▶ Convolution



Control cut

Proposed method

Concave



[ABCDCEDEBCDCDDDEE][BCDCDCDCEBCDCDCDCEBF]

W potential misses

W potential misses

Cut in two

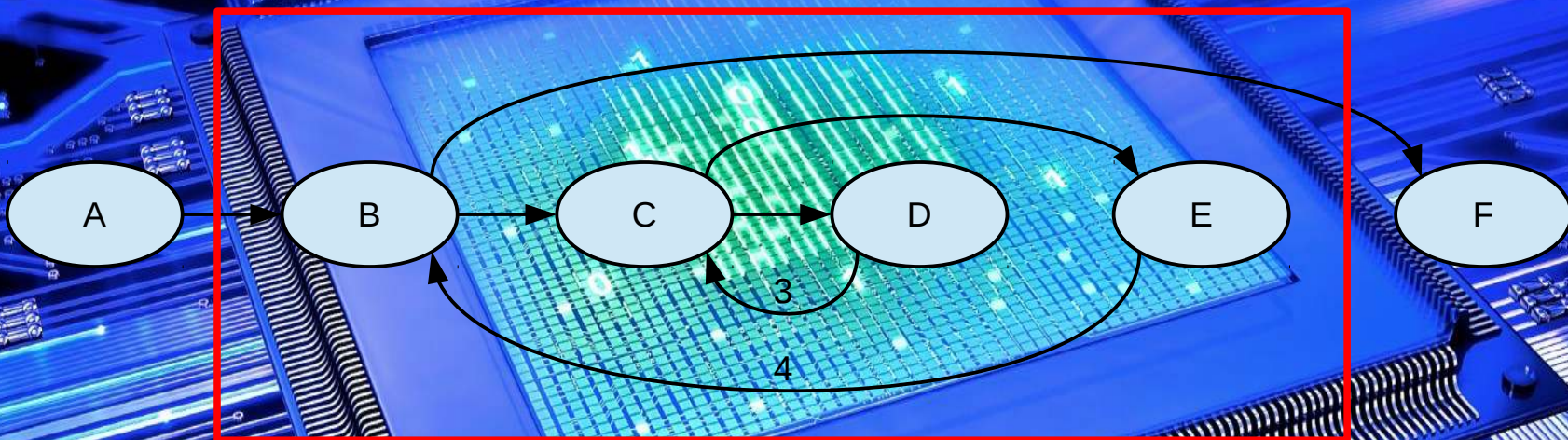
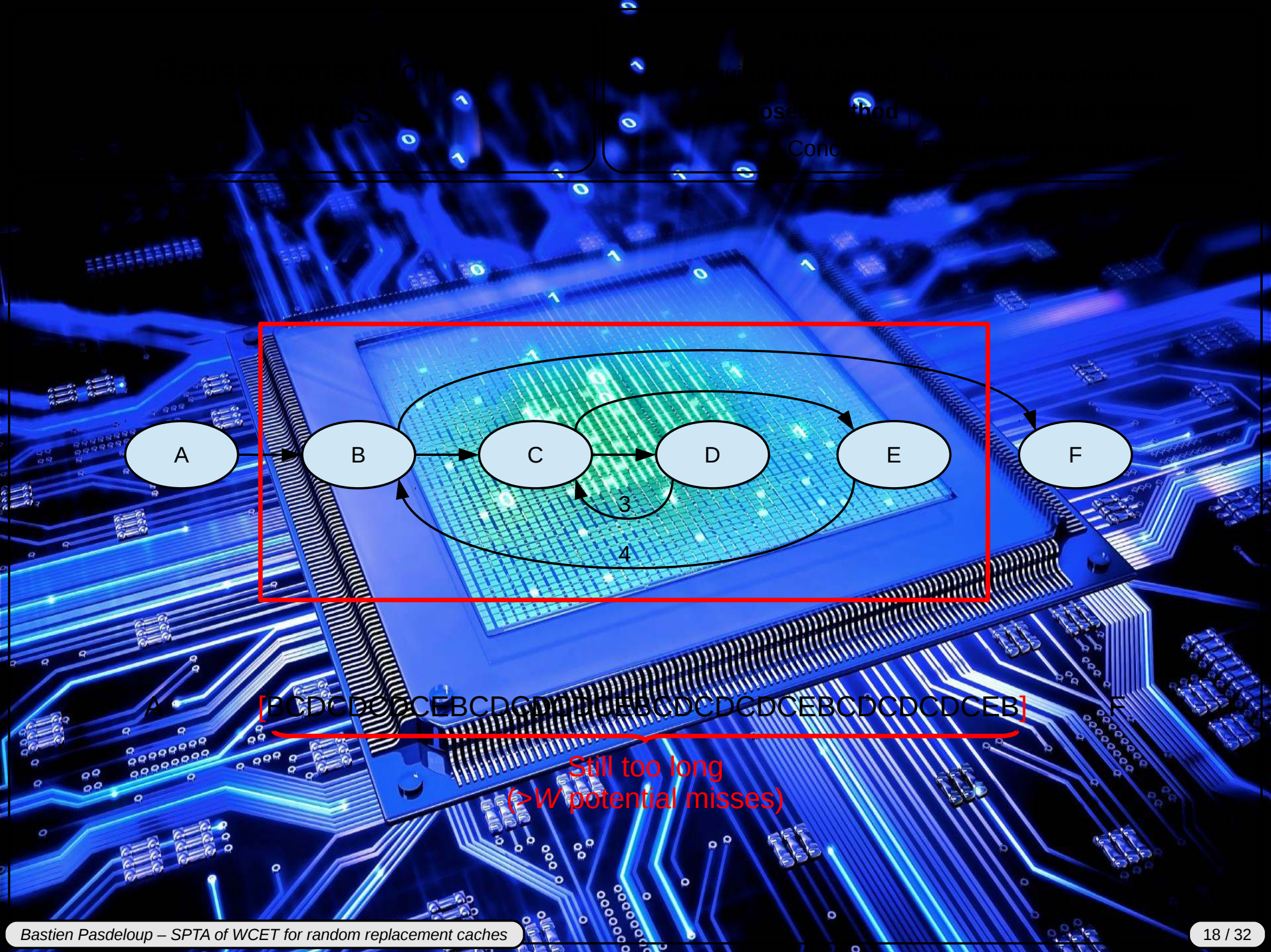




Safe cache

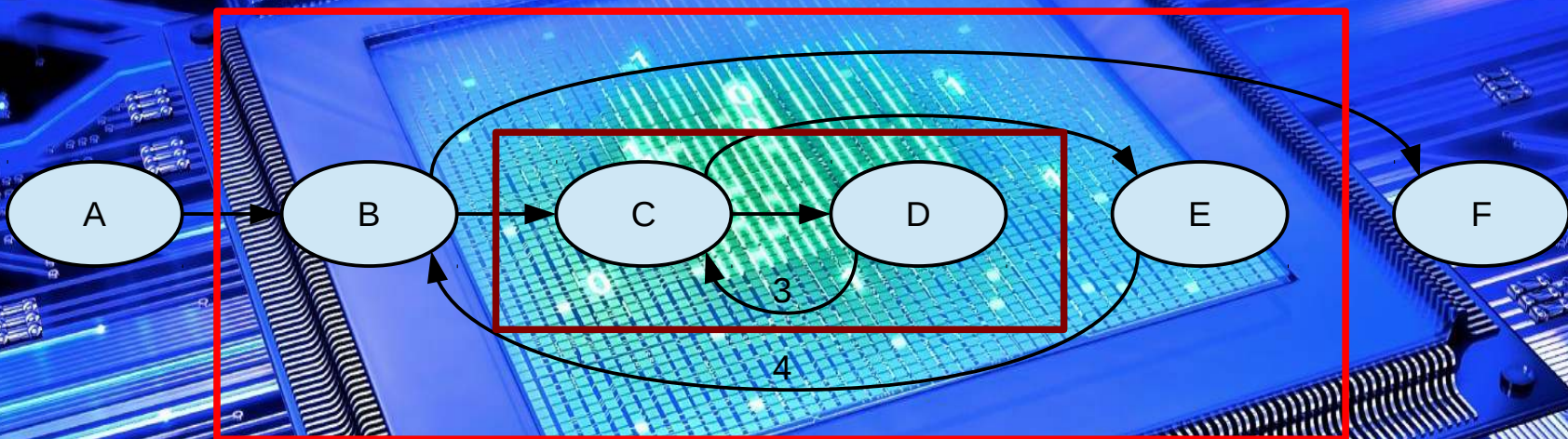
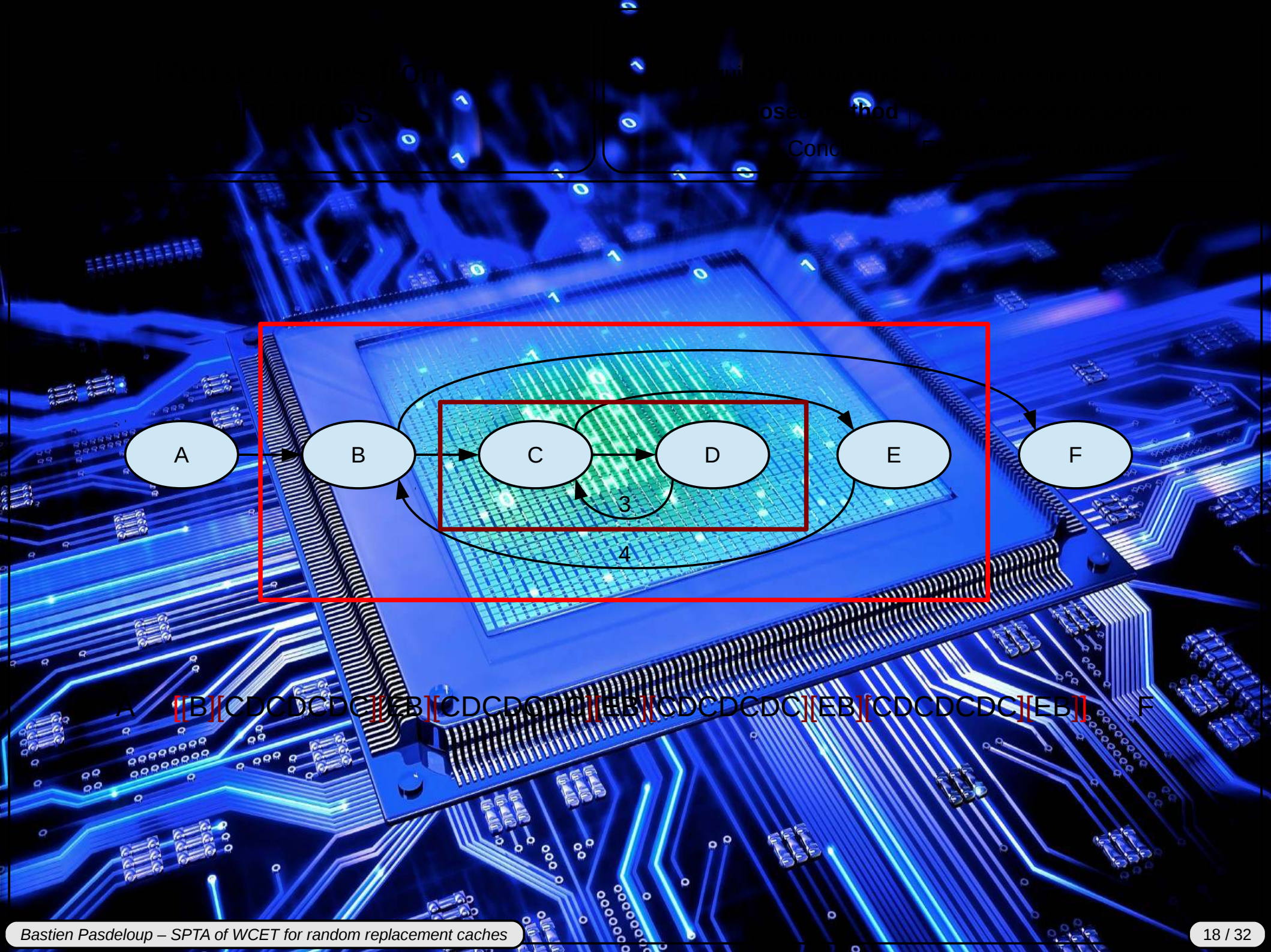




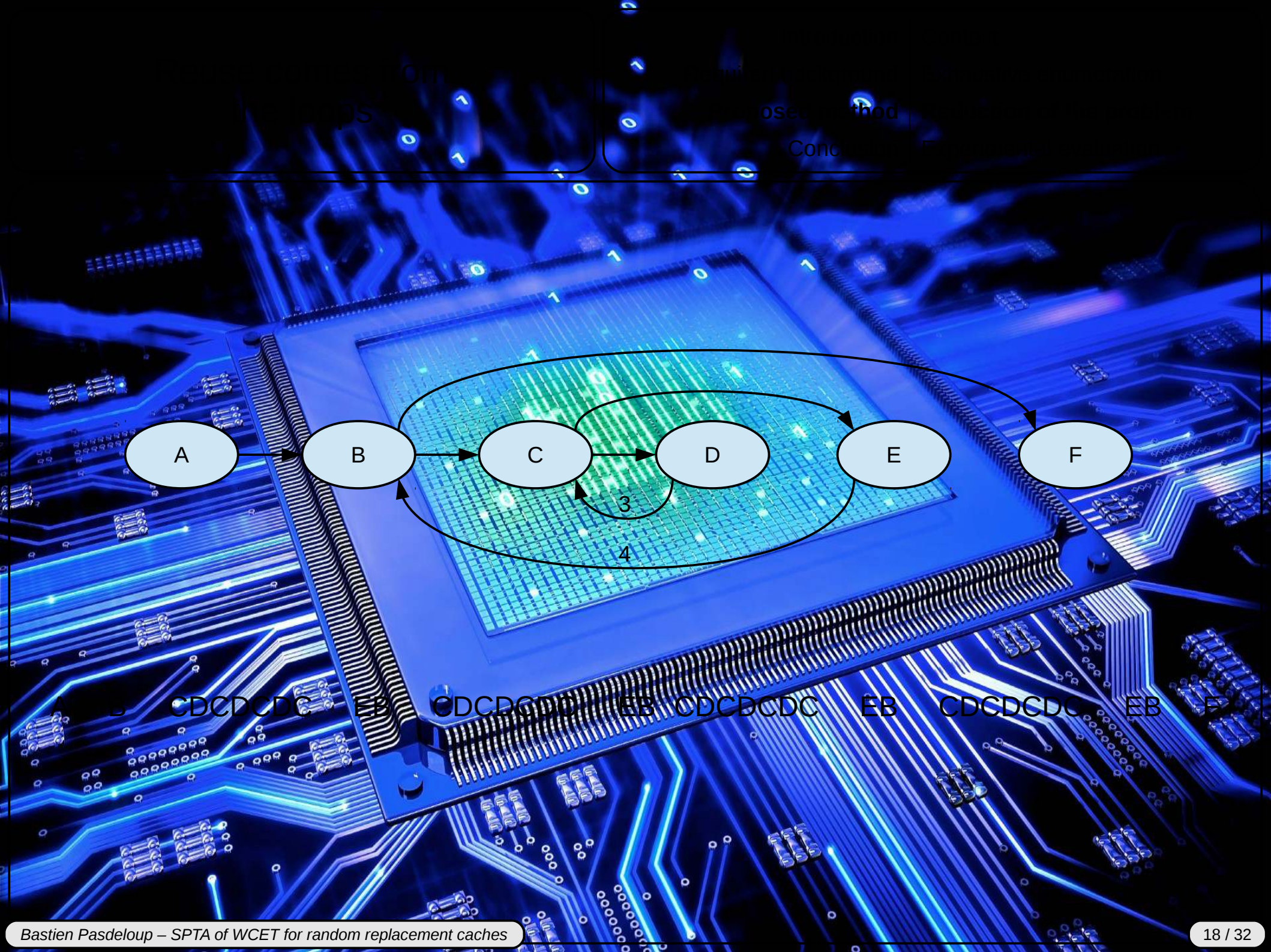


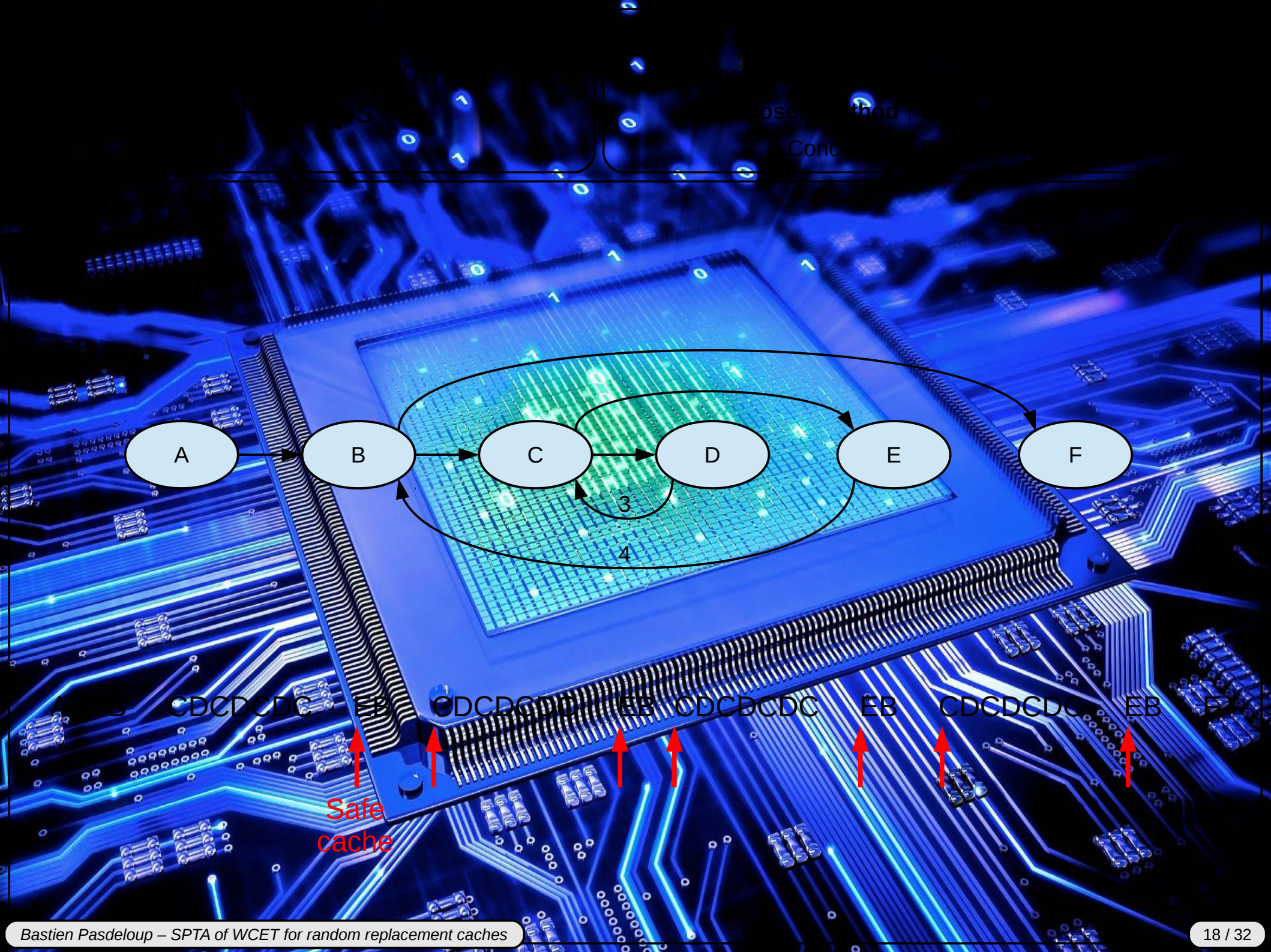
A [BCDCDCDCCEBCDCDGDDBCEBCDCDCDCDCCEBCDCDCDCDCB] F

Still too long
($> W$ potential misses)

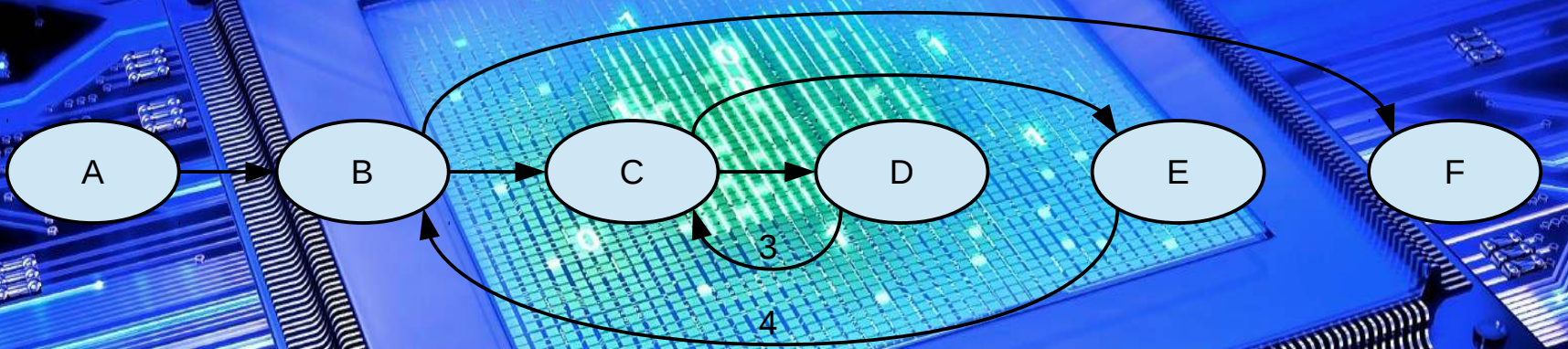


A **[[B]]CDCDCDC** **[[B]]CDCDCDC** **[[B]]CDCDCDC** **[[B]]CDCDCDC** **[[B]]CDCDCDC** **[[B]]CDCDCDC** F



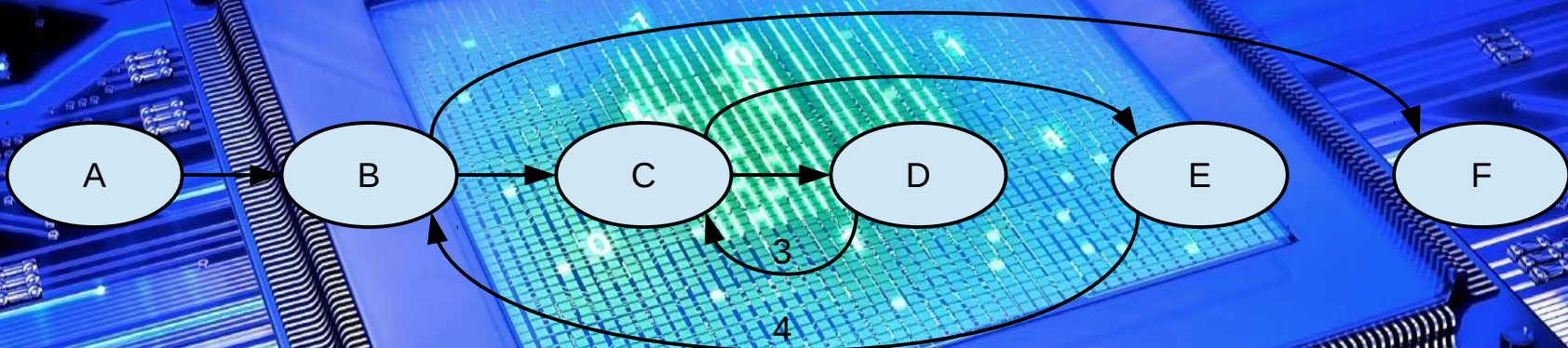


On the left: PS
On the right: Proposed method
Conclusion: The proposed method is more accurate than the PS method.



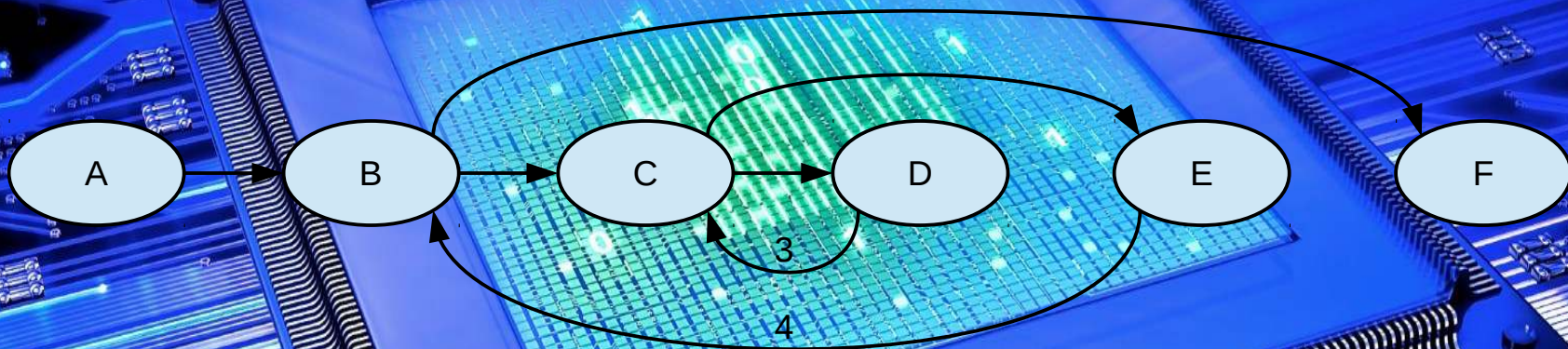
CDCDCDC EB CDCDCDC EB CDCDCDC EB CDCDCDC EB

Safe
cache

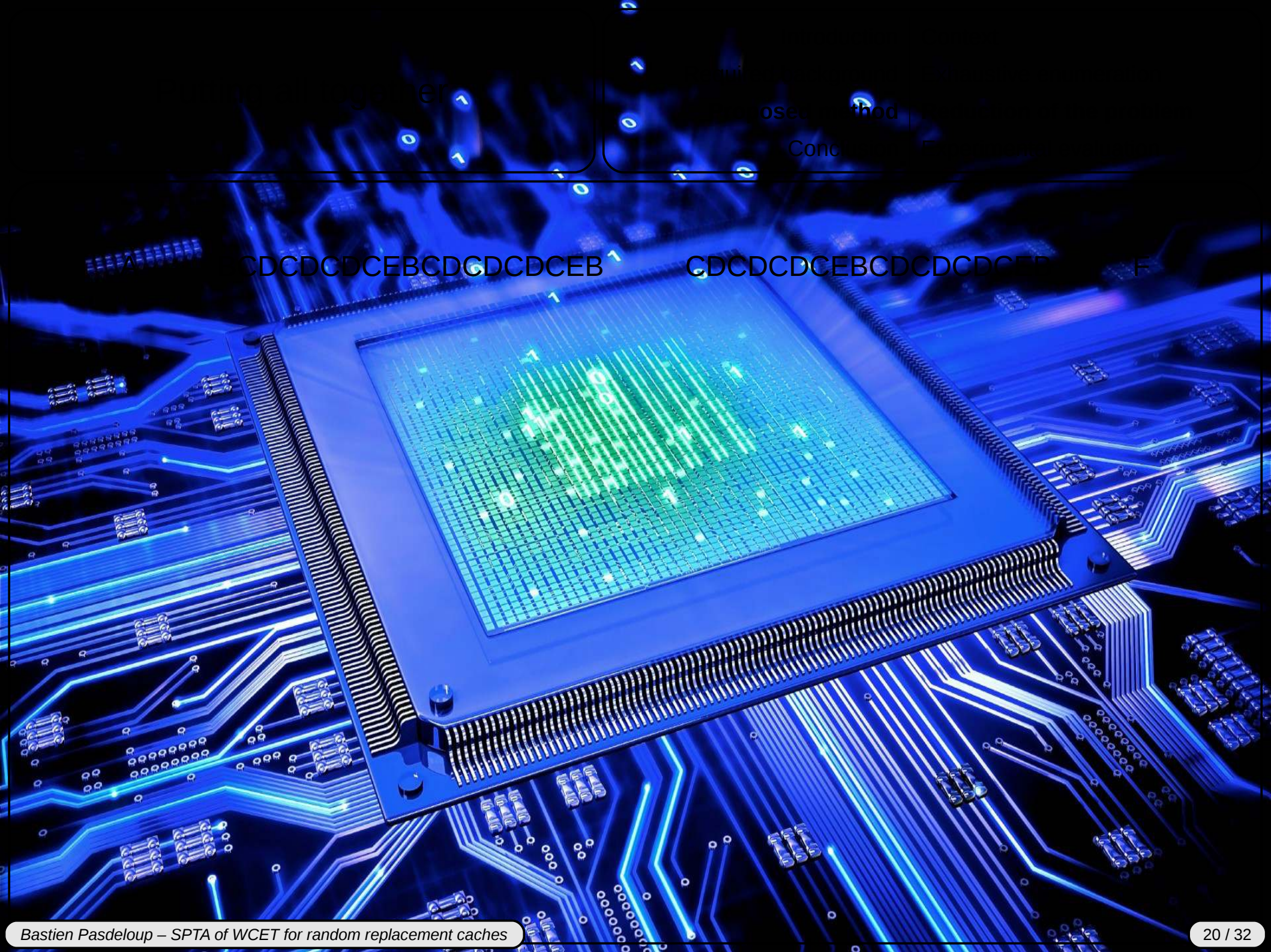


≤W potential misses

≤W potential misses



A B C D C D C D E B C D C D D C E B C D C D C D E B C D C D C D C D C E B



Agner

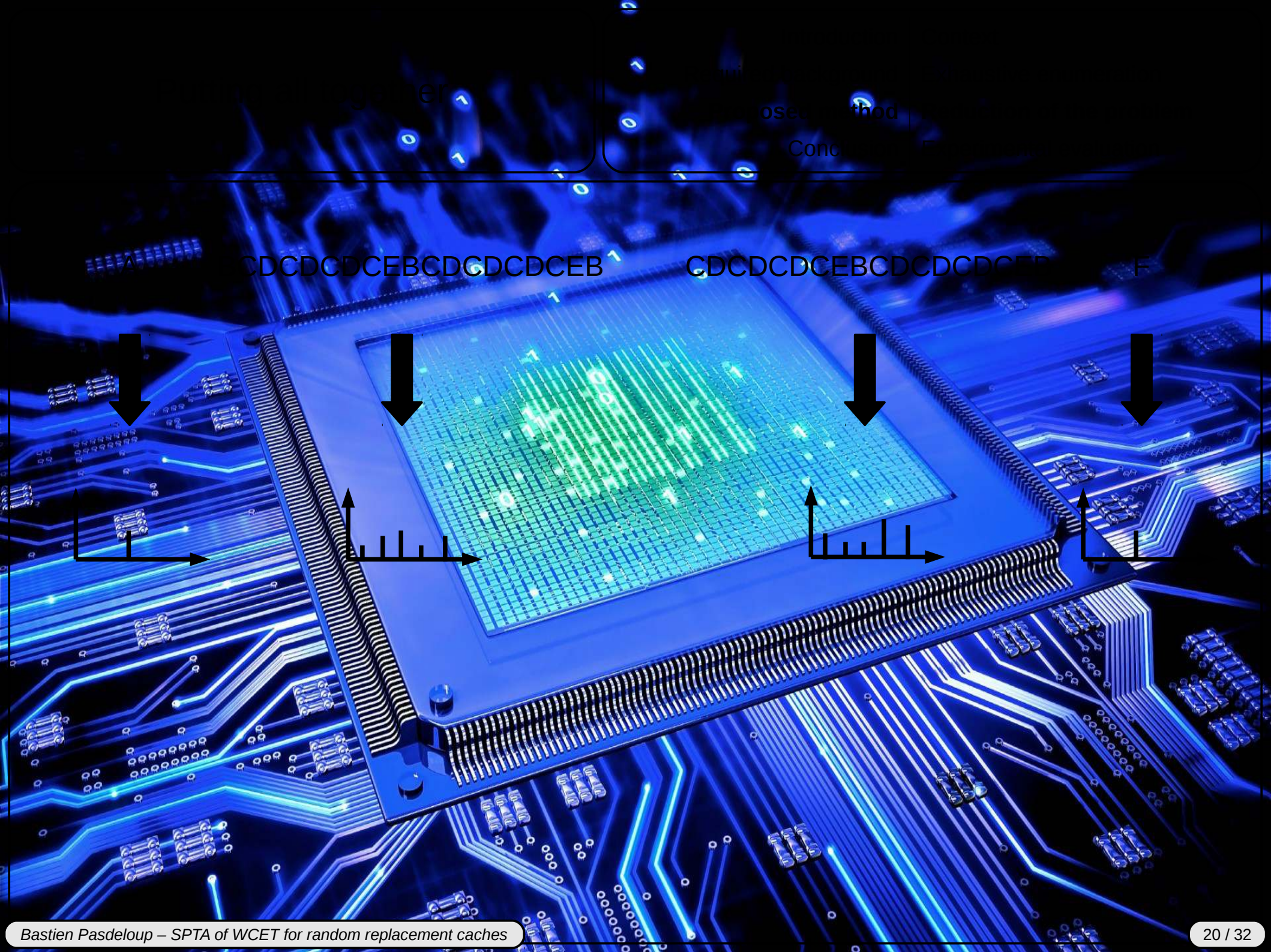
posed method

Concurrence

A CDCDCDCEBCDCDCDCB

CDCDCDCEBCDCDCDCB

F



Algorithm

Proposed method

Concave function

CDCDCDCEBCDCDCDCB

CDCDCDCEBCDCDCDCB

F

Aggregating

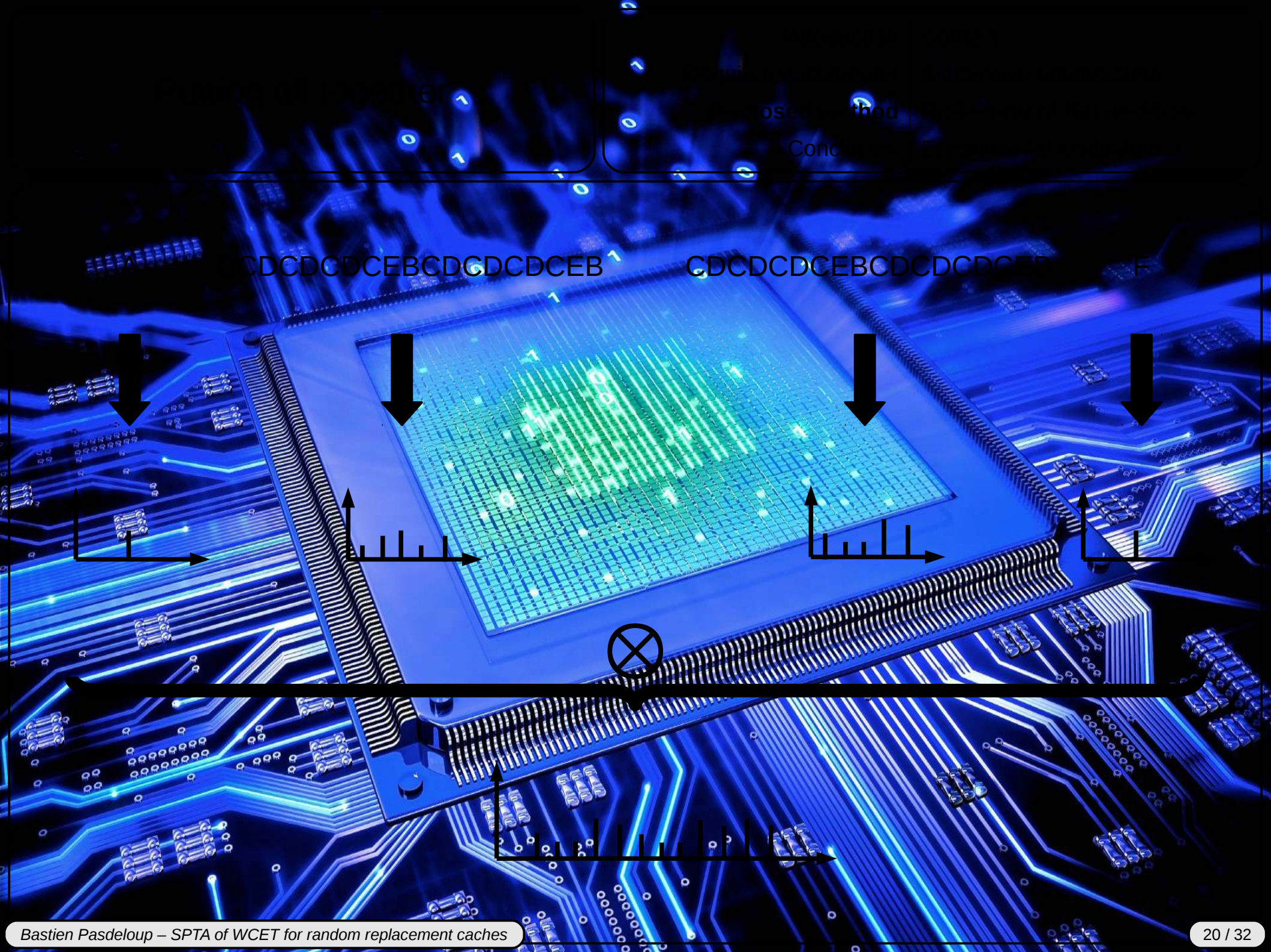
Proposed method

Concave-convex

A CDCDCDCEBCDCDCDCB

CDCDCDCEBCDCDCDCB

F



- ▶ Windows share references $\Leftrightarrow$ Precision loss
- ▶ Infinite-size windows $\Rightarrow$ Exact distribution

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- ▶ Infinite-size windows $\Rightarrow$ Exact distribution

Independence

- ▶ Same initial state for all analyses of windows
- ▶ Convolution can be applied

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- ▶ Infinite-size windows $\Rightarrow$ Exact distribution

Independence:

- ▶ Same initial state for all analyses of windows
- ▶ Convolution can be applied

Safety:

- ▶ Worst-case initial state
- ▶ Underestimation of hit probabilities
- ▶ Safe by construction

Aggressiveness

Proposed method

Concurrence

- ▶ Mälardalen benchmark suite
- ▶ 11 programs
- ▶ Single-path (or one trace if multi-paths)

Aggressiveness

Proposed method

Concurrence

- ▶ Mälardalen benchmark suite
- ▶ 11 programs
- ▶ Single-path (or one trace if multi-paths)

Cache:

- ▶ Set-associative (generalization of fully-associative)
- ▶ 8 sets, 4 ways, 32B lines (1KB cache)
- ▶ Modulo placement, random replacement

Aggressiveness

Proposed method

Concurrence Error

- ▶ Mälardalen benchmark suite
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Cache:

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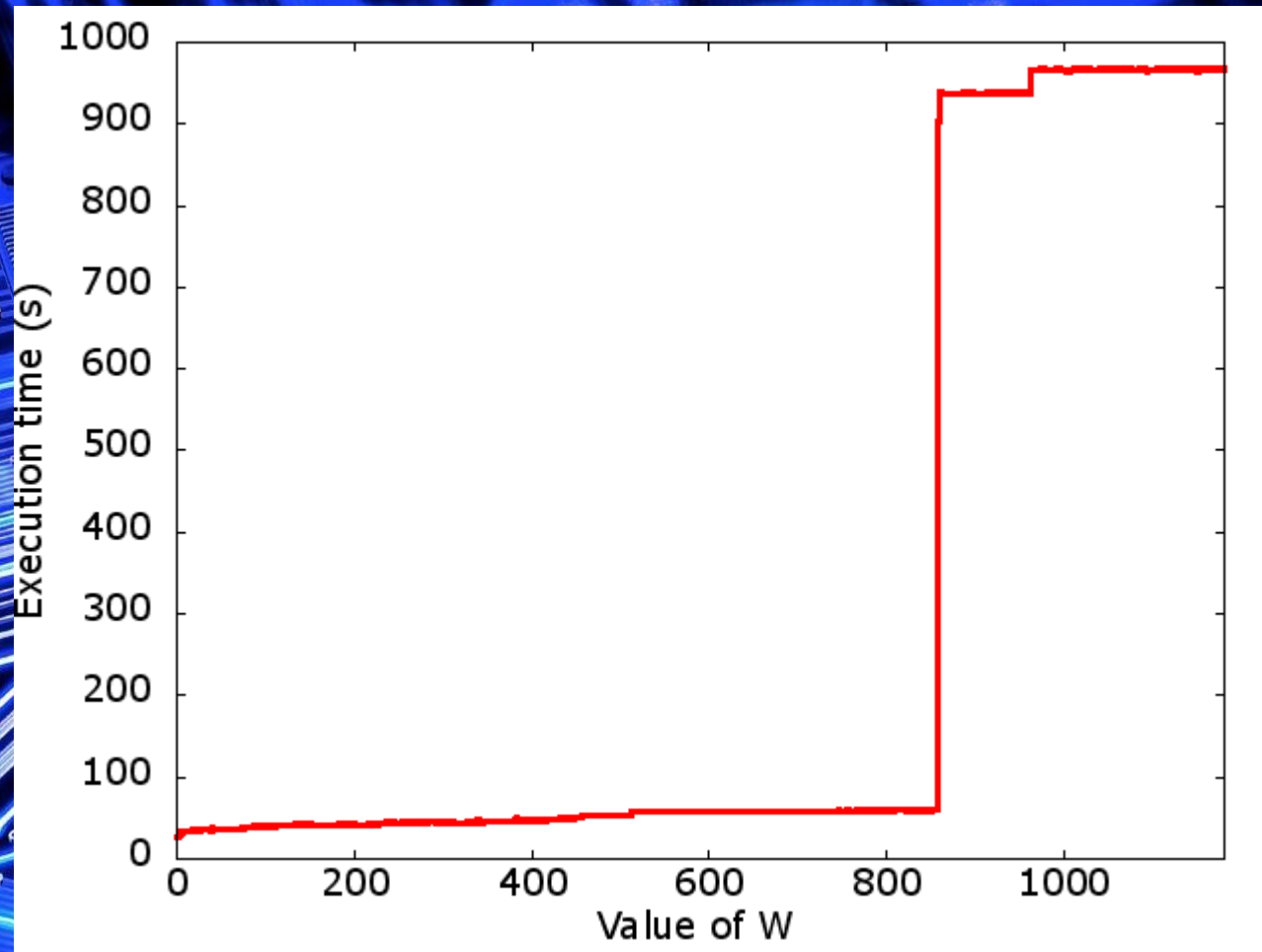
Approaches compared:

- ▶ Our approach (our implementation)
- ▶ Simulations (10000)
- ▶ Altmeyer & Davis [Alt2014] (~15mn max, always higher)

[Alt2014] Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis"

Execution time

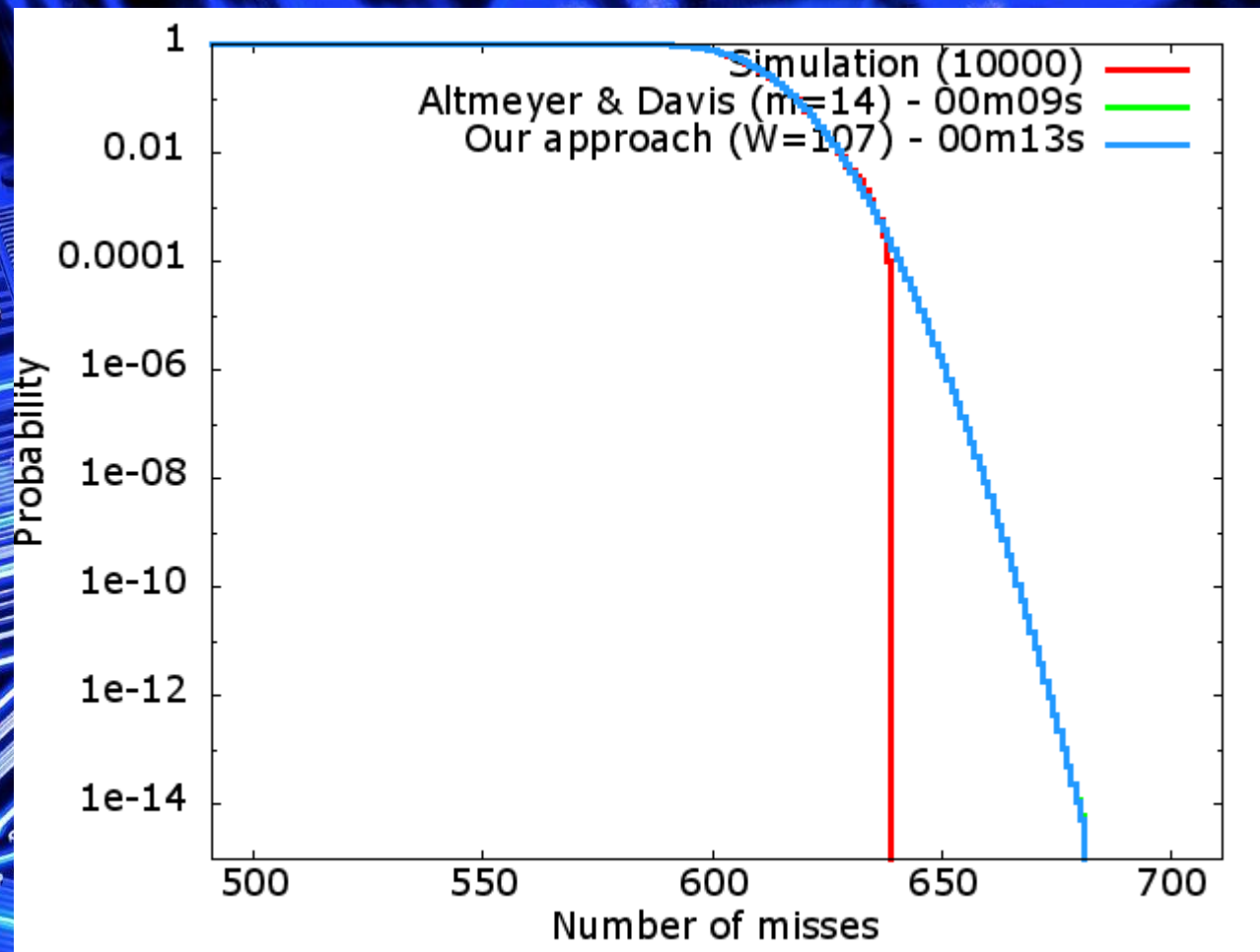
Proposed method
Concurrence Error



edn (39714 references)

Analysis
of the program

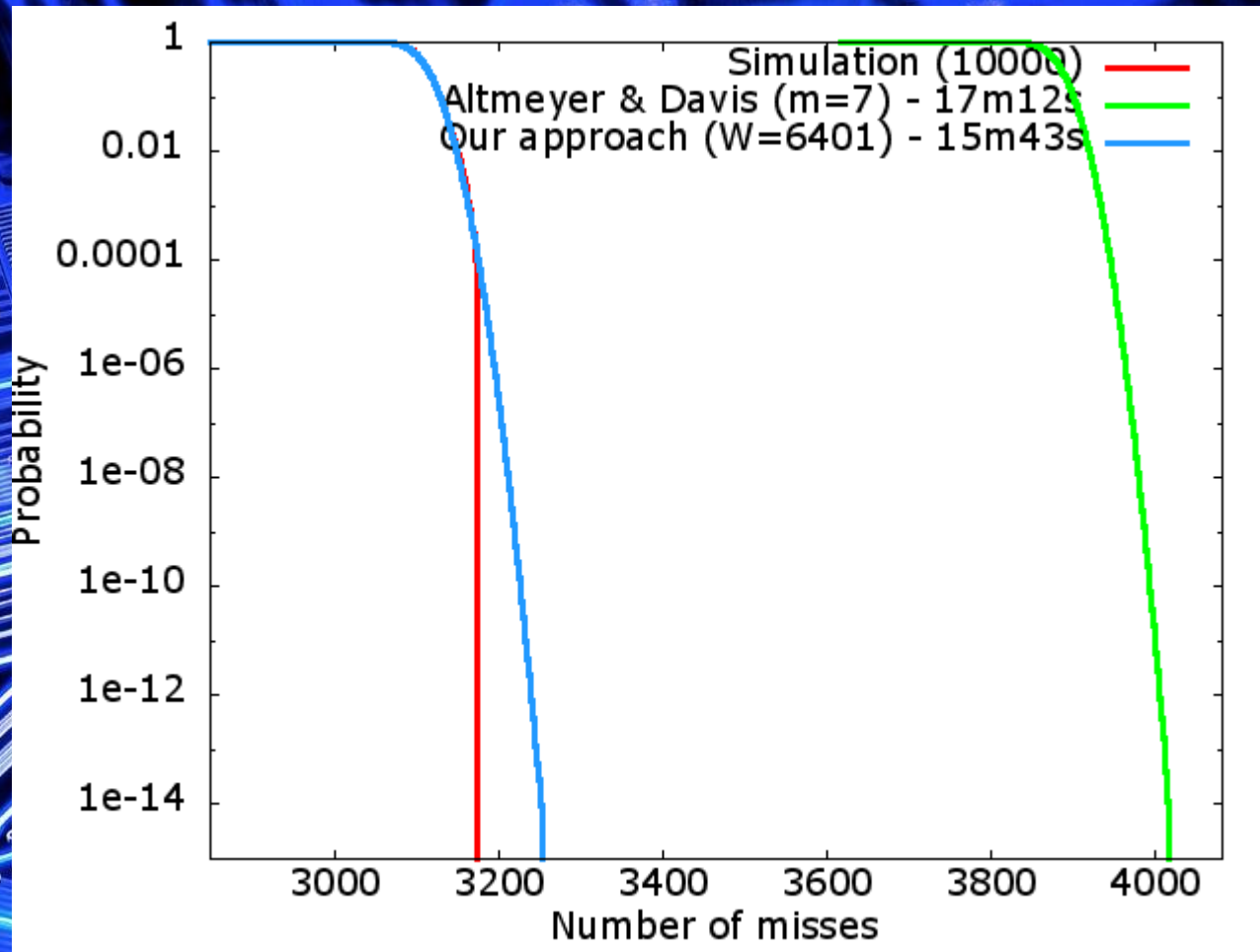
Simulation of the
proposed method
Conclude on the
effectiveness



Index (784 references)

Analysis
of the program

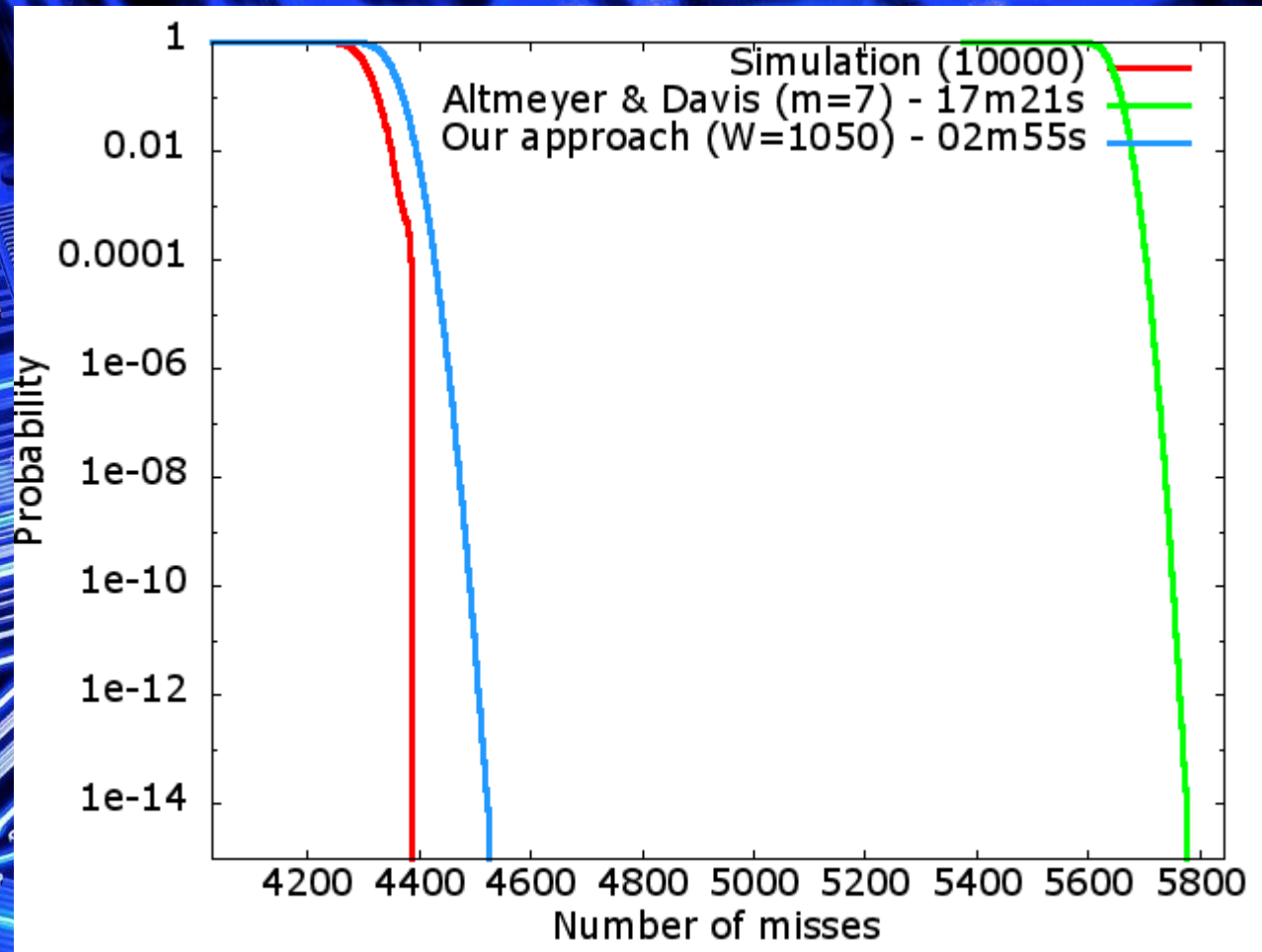
Simulation
of the proposed method
Conclusions



edn (39714 references)

Analysis
of program

Simulation
of proposed method
Conclude



fdct-jfdctint-minver-edn (83222 references)

Proposed Method

Conclusion

- ▶ Static
- ▶ Probabilistic
- ▶ Random aspects of the cache

- ▶ Static
- ▶ Probabilistic
- ▶ Random aspects of the cache

Applicability:

- ▶ Random replacement caches
- ▶ Single-path programs

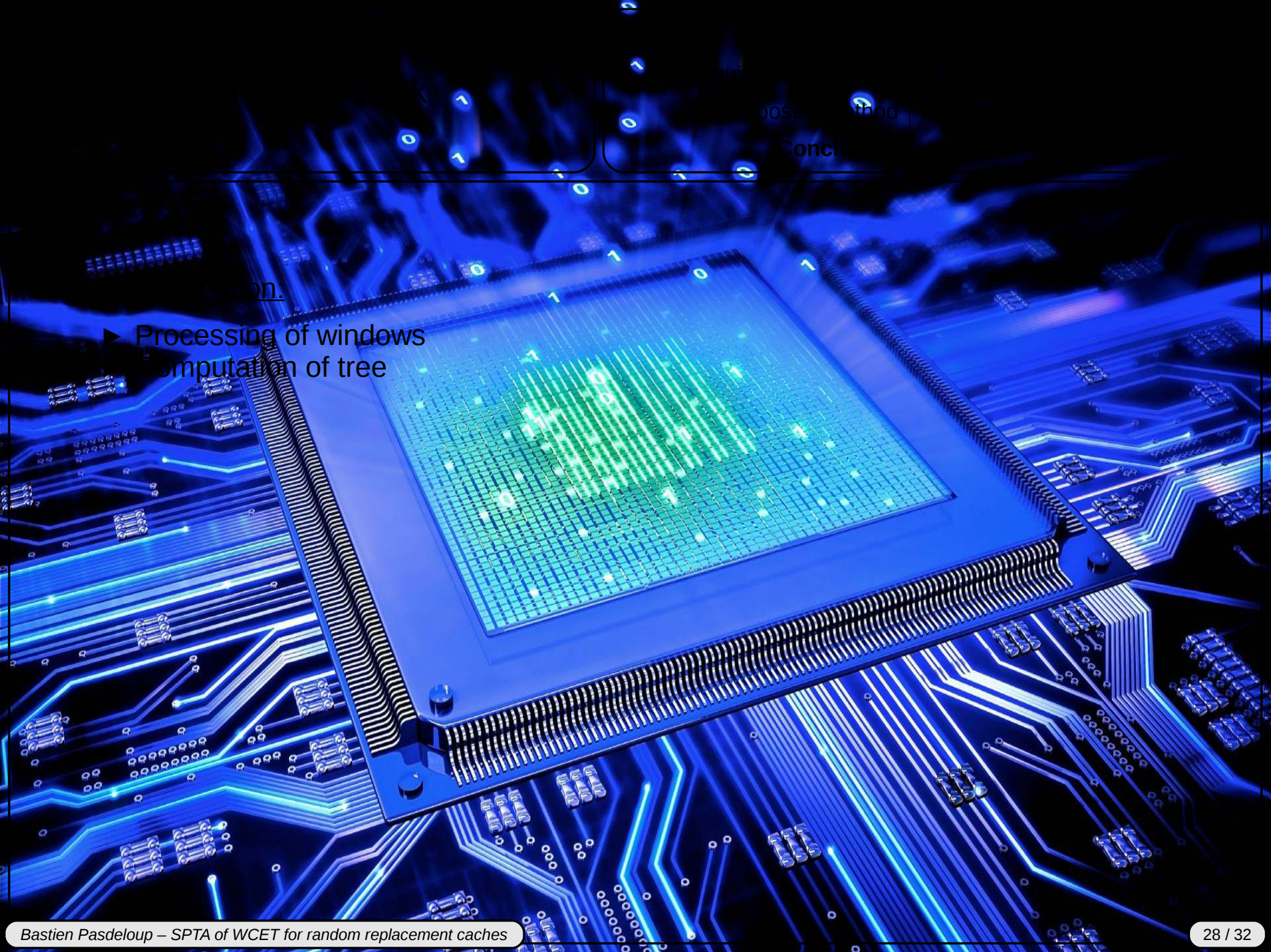
- ▶ Static
- ▶ Probabilistic
- ▶ Random aspects of the cache

Applicability:

- ▶ Random replacement caches
- ▶ Single-path programs

Important properties:

- ▶ Safe by construction
- ▶ Tightness experimentally shown



<

Choose method

Conclusion

Conclusion:

- Processing of windows
- Computation of tree



Goal

Proposed method

Conclusion

Conclusion:

- Processing of windows
- Computation of tree

Combination of approaches:

- Partition into windows
- Altmeyer & Davis [Alt2014]

[Alt2014] Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis"

Proposed method

Conclusion

Extension:

- ▶ Processing of windows
- ▶ Computation of tree

Combination of approaches:

- ▶ Partition into windows
- ▶ Altmeyer & Davis [Alt2014]

Extension to multi-path programs

[Alt2014] Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Tiling Analysis"

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Proposed method
Conclusion

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Proposed method

Conclusion

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Static probabilistic timing analysis of worst-case execution time for random replacement caches

Student: Bastien Pасdeloup

Supervisors: Isabelle Puaut, Damien Hardy

Research team: ALF

Wednesday, June 26th 2014